18.917: The Chromatic Splitting Conjecture

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Contents

How strange to actually have to see the path of your journey in order to make it.

—Neal Shusterman, [Shu16]

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Theme 1
Nishida Nilpotence

1.1 February 2

Here we go.

1.1.1 Idempotent Algebras

The goal of this class is to understand some topics related to the chromatic splitting conjecture. Thus, the first half of the class will try to understand the statement, and the second half of the class will explain how it relates to other problems in algebra.

Warning 1.1. All categories in this course are ∞-categories.

Example 1.2. Given a ring R, we have a stable, symmetric monoidal ∞-category $D(R)$ of chain complexes of R-modules, considered up to quasi-isomorphism. Notably, the symmetric monoidal structure is given by the derived tensor product.

We begin our story with idempotent algebras.

Definition 1.3 (idempotent algebra)**.** Fix a ring R. An *idempotent algebra* is an object $E \in D(R)$ equipped with a unit map $R \to E$ such that the composite

$$E = E \otimes_R R \to E \otimes_R E$$

is an equivalence.

Remark 1.4. Such an object E grants E a multiplication structure $E \otimes_R E \to E$, and E gains the structure of a differentially graded algebra.

Example 1.5. Consider $R = \mathbb{Z}$. Then for each prime p, the algebra $\mathbb{Z}_{(p)}$ is idempotent: localizing $\mathbb{Z}_{(p)}$ further at (p) does nothing!

Non-Example 1.6. The $\mathbb{Z}$-algebra $\mathbb{F}_p$ is not idempotent because the tensor product we are considering is derived. Indeed, we computed $\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p$ last semester.

Here is a quick reason why one might care about idempotent algebras.

Theorem 1.7 (Neeman)**.** Fix a Noetherian ring R. Then the lattice of idempotent algebras is equivalent to the data of $\operatorname{Spec} R$ as a topological space.

Example 1.8. For $R = \mathbb{Z}$, it turns out that the idempotent algebras are either $\mathbb{Z}_{(p)}$ or $\mathbb{Q}$, and the maps between them look like the specializations of $\operatorname{Spec} R$.

Of course, we are homotopy theorists, so we have less reason to care about $\mathbb{Z}$. Recall that $\mathbb{Z}$ is obtained from $\mathbb{N}$ by formally adding inverses. But $\mathbb{N}$ is basically isomorphism classes of FinSet; if we had instead formally added inverses directly to FinSet (instead of taking isomorphism classes first), we would have found the sphere spectrum $\mathbb{S}$. In particular, we will be interested in the category $D(\mathbb{S})$ of $\mathbb{S}$-modules, also called spectra.

We now no longer have access to algebraic geometry directly on $\mathbb{S}$. Instead, Theorem 1.7 motivates us to look for the idempotent algebras for $\mathbb{S}$.

Remark 1.9. For any $x \in \pi_*\mathbb{S}$, there is an idempotent algebra $\mathbb{S}\left[x^{-1}\right]$. For example, $\pi_0\mathbb{S} = \mathbb{Z}$, so there is an idempotent algebra $\mathbb{S}_{(p)}$.

Here is our first main theorem.

Theorem 1.10 (Nishida)**.** Fix some $x \in \pi_*\mathbb{S}$ of positive degree. Then x is nilpotent.

Thus, the idempotent algebras $\mathbb{S}\left[x^{-1}\right]$ do not look genuinely "new." To get other idempotent algebras, we need more tools.

1.1.2 The Adams–Novikov Spectral Sequence

Recall the $\mathbb{S}$-algebra MU defined as the colimit of the embedding $\mathrm{BU} \to \mathrm{BLG}(\mathbb{S}) \subseteq \mathrm{Mod}(\mathbb{S})$. Let's compute its homotopy groups.

Definition 1.11 (formal group law)**.** Fix a commutative ring R. Then a *commutative formal group law* over R is a power series $f(x, y) \in R[[x, y]]$ satisfying

(a) $f(x, 0) = x$ and $f(0, y) = y$,

(b) $f(x, y) = f(y, x)$, and

(c) $f(x, f(y, z)) = f(x, f(y, z))$.

Definition 1.12 (Lazard ring)**.** The *Lazard ring* is the ring L which is exactly the quotient of $\mathbb{Z}[\{a_{ij}\}_{ij}]$ by the relations dictating that

$$f(x, y) = \sum_{i,j \geq 0} a_{ij} x^i y^j$$

is a commutative formal group law.

Remark 1.13. In other words, L represents the collection formal group laws, in the sense that the data of a formal group law for a ring R amounts to the data of a ring homomorphism $L \to R$.

Remark 1.14. By definition, there is a "universal" formal group law f_L in L given exactly by

$$f_L(x,y) = \sum_{i,j\geq 0} a_{ij}x^i y^j.$$

Theorem 1.15 (Quillen)**.** The ring $\pi_*\mathrm{MU}$ is exactly the Lazard ring.

Remark 1.16. Quillen also computed $\pi_*(\mathrm{MU}\otimes_{\mathbb{S}}\mathrm{MU})$ as well as the two natural maps $\pi_*\mathrm{MU} \to \pi_*(\mathrm{MU}\otimes_{\mathbb{S}}\mathrm{MU})$. It turns out that this is more or less related to some notion of isomorphism of the formal group laws.

The use of MU is that it produces a spectral sequence with which we can understand $\pi_*\mathbb{S}$. By Čech descent along the map $\mathbb{S} \to \mathrm{MU}$, we see that $\mathbb{S}$ is the limit of the diagram

$$\mathrm{MU} \rightrightarrows \mathrm{MU}\otimes_{\mathbb{S}}\mathrm{MU} \substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow} \mathrm{MU}\otimes_{\mathbb{S}}\mathrm{MU}\otimes_{\mathbb{S}}\mathrm{MU} \qquad \cdots$$

which we can then truncate as $\mathrm{fil}^n\,\mathbb{S}$ in order to get a descending filtration to $\mathrm{fil}^0\,\mathbb{S}$. Computing homotopy along this filtration produces the desired spectral sequence, as soon as we compute homotopy groups of the various tensor powers of the MUs and so on.

Theorem 1.17 (Adams–Novikov)**.** Let $\mathcal{M}_{\mathrm{fg}}$ be the moduli space of formal groups. Then there is a spectral sequence

$$E_2 = \mathrm{H}^s(\mathcal{M}_{\mathrm{fg}};\omega^{\otimes t}) \Rightarrow \pi_{2t-s}\mathbb{S}.$$

Remark 1.18. It turns out that the spectral sequence is concentrated in the region $s \leq 2t-s$.

Example 1.19. Along the line $s = 2t-s$, there is some h_1 at $(s, 2t-s) = (1,1)$, and then we can take powers of it to go up the line. It turns out that h_1 survives the spectral sequence, and it goes to the "Hopf map" $\eta \in \pi_1\mathbb{S}$; however, $\eta^4 = 0$, though the Adams–Novikov spectral sequence cannot see it!

Thus, we see that the Adams–Novikov spectral sequence is not an amazing approximation: the E_2 page sees many classes which we know abstractly must vanish! Life is better if we pass to E_∞ instead; the following is our first main theorem.

Theorem 1.20 (Devinatz–Hopkins–Smith)**.** The E_∞ page of the Adams–Novikov spectral sequence lies under a curve which grows more slowly than any line.

Note that this immediately implies Theorem 1.10. On the other hand, we will see that the topological input of Theorem 1.10 plus some algebraic facts about formal group laws will prove the above big theorem.

Remark 1.21. The curve is known to be faster than logarithmic, but not much else is known. Our proof will not help us much because our proof of Theorem 1.10 will be ineffective.

1.1.3 Back to Idempotent Algebras

Let's return to trying to find some idempotent algebras.

Notation 1.22. Define the power series $[n] \in L[[x]]$ to be adding with f a total of n times.

Example 1.23. We see that $[2](x) = f(x,x)$ and $[5](x) = f(f(f(f(x,x),x),x),x)$.

Notation 1.24. Fix a prime p. Then we define the class $v_n \in \pi_* \mathrm{MU}$ to be the coefficient of x^{p^n} in the power series $[p](x)$.

Now, because localization is exact, we see that $\mathbb{S}_{(p)}$ is the limit of the nerve

$$\mathrm{MU}_{(p)} \Longleftarrow\!\!\!\!\Rightarrow \mathrm{MU}_{(p)} \otimes_{\mathbb{S}} \mathrm{MU}_{(p)} \Lleftarrow\!\!\!\!\Rrightarrow \mathrm{MU}_{(p)} \otimes_{\mathbb{S}} \mathrm{MU}_{(p)} \otimes_{\mathbb{S}} \mathrm{MU}_{(p)} \qquad \cdots$$

so it is not unreasonable to consider the following limit.

Notation 1.25. Fix a prime p and some $n \geq 0$. Then we define $L_n\mathbb{S}_{(p)}$ as the limit of the following diagram.

$$\mathrm{MU}_{(p)}\left[v_n^{-1}\right] \Longleftarrow\!\!\!\!\Rightarrow \mathrm{MU}_{(p)}\left[v_n^{-1}\right] \otimes_{\mathbb{S}} \mathrm{MU}_{(p)}\left[v_n^{-1}\right] \Lleftarrow\!\!\!\!\Rrightarrow \mathrm{MU}_{(p)}\left[v_n^{-1}\right] \otimes_{\mathbb{S}} \mathrm{MU}_{(p)}\left[v_n^{-1}\right] \otimes_{\mathbb{S}} \mathrm{MU}_{(p)}\left[v_n^{-1}\right]$$

We may abbreviate $L_n\mathbb{S}_{(p)}$ to $L_n\mathbb{S}$ if there is no possibility of confusion.

Remark 1.26. It turns out that there are natural maps $L_{n+1}\mathbb{S} \to L_n\mathbb{S}$.

These spectra $L_n\mathbb{S}$ give us new idempotent algebras, more or less granting us further understanding of the "spectrum" of $\mathbb{S}$.

Theorem 1.27 (Hopkins–Ravenel)**.** Fix a prime p and some $n \geq 0$. Then $L_n\mathbb{S}_{(p)}$ is an idempotent algebra.

Remark 1.28. Ravenel has conjectured that if E is a nonzero idempotent algebra under $\mathbb{S}_{(p)}$, then E is either $\mathbb{Q}$ or one of the $L_n\mathbb{S}$s. This was recently disproved. It is current work to attempt a classification.

Nonetheless, $\mathbb{S}_{(p)}$ can be understood well from the $L_n\mathbb{S}$s.

Theorem 1.29 (Hopkins–Ravenel)**.** Fix a prime p. Then $\mathbb{S}_{(p)}$ is the limit of the diagram

$$\cdots \to L_3\mathbb{S}_{(p)} \to L_2\mathbb{S}_{(p)} \to L_1\mathbb{S}_{(p)}.$$

1.1.4 Completion

Continue with our fixed prime p. For motivation, we return to abelian groups.

Remark 1.30. For any $M \in D(\mathbb{Z})$, the p-localization sits in a pullback square

$$\begin{tikzcd} M_{(p)} \arrow[r] \arrow[d] & M_p^\wedge \arrow[d] \\ M \otimes_{\mathbb{Z}} \mathbb{Q} \arrow[r] & M_p^\wedge \otimes_{\mathbb{Z}} \mathbb{Q} \end{tikzcd}$$

more or less corresponding to finding the "lattice" $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$.

Analogously, there is a completion of $L_n\mathbb{S}$ which fits into a diagram

$$\begin{tikzcd} L_nE \arrow[r] \arrow[d] & L_{K(n)}E \arrow[d] \\ L_{n-1}E \arrow[r] & L_{n-1}L_{K(n)}E \end{tikzcd} \tag{1.1}$$

where $L_nE := L_n\mathbb{S} \otimes_{\mathbb{S}} E$. We are now ready to state the chromatic splitting conjecture.

Conjecture 1.31 (Chromatic splitting)**.** For any $n \geq 2$, the inclusion

$$L_{K(n)}\mathbb{S} \to L_{n-1}L_{K(n)}\mathbb{S}$$

is an inclusion of a direct summand.

Remark 1.32. This implies that the natural map

$$\mathbb{S}_p^\wedge \to \prod_{n\geq 1} L_{K(n)}\mathbb{S}$$

is the inclusion of a direct summand. The point is that the squares (1.1) are rather degenerate, which would let us compute the homotopy groups of $L_n\mathbb{S}$ from the completions.

Remark 1.33. Conjecture 1.31 is known at $n = 2$ and all primes, by work of many people.

The goal of the present class is to review the homotopy theory required to understand the statement of Conjecture 1.31 formally, and then we will discuss why perfectoid geometry may be useful to prove it.

Let's see why passing to $L_{K(n)}\mathbb{S}$ is genuinely easier.

Example 1.34. For $p > 2$, we can define $L_{K(1)}\mathbb{S}$ as the homotopy fiber of the endomorphism $\psi^g - 1$ of $\mathrm{KU}_p^\wedge$, where g is a choice of topological generator of $\mathbb{Z}_p^\times$, and ψ is some action of $\mathbb{Z}_p^\times$ on $\mathrm{KU}_p^\wedge$.

Theorem 1.35 (Goerss–Hopkins–Miller, Rogres)**.** Fix a prime p. For each $n \geq 1$, there is an $\mathbb{S}$-algebra E_n and a profinite group $\mathbb{G}_n$ for which

$$L_{K(n)}\mathbb{S} = (E_n)^{\mathbb{G}_n}.$$

In fact, E_n is a Galois extension of $L_{K(n)}\mathbb{S}$.

Remark 1.36. We will only be able to keep track of this sort of "infinite Galois theory" with condensed mathematics.

Remark 1.37. The profinite group $\mathbb{G}_n$ is some subgroup of automorphisms of formal group laws.

Remark 1.38. For any spectrum X, there is some "Galois descent"

$$L_{K(n)}X = \left(L_{K(n)}(E_n \otimes X)\right)^{\mathbb{G}_n}.$$

This generalizes to a spectral sequence

$$\mathrm{H}^*_{\mathrm{cts}}(\mathbb{G}_n; \pi_*(L_{K(n)}(E_n \otimes X))) \Rightarrow \pi_* L_{K(n)}X.$$

The previous remark produces a spectral sequence

$$\mathrm{H}^*_{\mathrm{cts}}(\mathbb{G}_n; \pi_* E_n) \Rightarrow \pi_* L_{K(n)}\mathbb{S}.$$

If p is large compared to n, then it turns out that the spectral sequence collapses for degree reasons, so we are reduced to a pure algebra problem.

The end of the course will be interested $L_{K(n-1)}L_{K(n)}\mathbb{S}_{(p)}$ for general n but p very large. Conjecture 1.31 tells us that this should be fairly easy to understand, so we can view the end of the course as trying to provide some evidence for the conjecture. For example, work in progress by many people has recently culminated in the following strategy.

Notation 1.39. Fix $\mathbb{B} := E_{n-1} \otimes_{\mathbb{S}} L_{K(n-1)}E_n$.

Remark 1.40. It turns out that $\mathbb{B}$ is Galois over $L_{K(n-1)}L_{K(n)}\mathbb{S}$ with Galois group $\mathbb{G}_{n-1} \times \mathbb{G}_n$. Thus, we can hope to be able to use some Galois descent spectral sequence to understand $L_{K(n-1)}L_{K(n)}\mathbb{S}$, as in Remark 1.38.

Now, $\pi_*\mathbb{B}$ is a local ring, so one becomes motivated to consider a perfection $\widehat{\mathbb{B}}$. In particular, it turns out that there is a $\mathbb{G}_n \times \mathbb{G}_{n-1}$-equivariant map $\mathbb{B} \to \widehat{\mathbb{B}}$, so taking fixed points produces a map out of $L_{K(n-1)}L_{K(n)}\mathbb{S}$. This is the sort of thing that Conjecture 1.31 asks us to do! Of course, the target is related to the perfection $\widehat{\mathbb{B}}$, which we now want to understand.

Theorem 1.41. The groups $\mathrm{H}^*_{\mathrm{cts}}(\mathbb{G}_n \times \mathbb{G}_{n-1}; \pi_*\widehat{\mathbb{B}})$ is the same as the cohomology of the structure sheaf of some diamond related to the Fargues–Fontaine curve.

Let's explain the application to Conjecture 1.31: this calculation tells us that $(\widehat{\mathbb{B}})^{\mathbb{G}_n \times \mathbb{G}_{n-1}}$ is $L_{K(n-1)}\mathbb{S} \oplus \Sigma L_{K(n-1)}\mathbb{S}$, from which our small piece of Conjecture 1.31 follows!

1.2 February 4

Our next goal is to prove Theorem 1.10.

1.2.1 Spectra

Let's set some notation. We are interested in the category Spaces of spaces, also called anima.

Definition 1.42 (pointed space)**.** The category of pointed spaces is denoted Space_*.

Remark 1.43. There is a functor $(-)_+ \colon \mathrm{Spaces} \to \mathrm{Spaces}_*$ which simply adds a basepoint to any topological space.

Our next category will be spectra.

Definition 1.44 (spectra)**.** A *spectrum* is an infinite tuple $(X_0, X_1, \ldots)$ of spaces, equipped with isomorphisms $X_0 \cong \Omega X_1 \cong \Omega^2 X_1 \cong \cdots$. The category of spectra is, unsurprisingly, denoted $\mathrm{Spectra}$.

Remark 1.45. The functor $\Omega \colon \mathrm{Spectra} \to \mathrm{Spectra}$, given by shifting all the spaces down, is an autoequivalence. Its inverse functor is Σ. Thus, $\mathrm{Spectra}$ is a stable category.

Remark 1.46. The functor $\Omega^\infty : \mathrm{Spectra} \to \mathrm{Spaces}$ given by sending a spectrum X to X_0 admits an adjoint Σ^∞. We may denote the composite functor $X \mapsto \Sigma^\infty(X_+)$ by Σ^∞_+.

Remark 1.47. The category $\mathrm{Spectra}$ admits products and coproducts, which are the same and denoted $\oplus$.

Example 1.48. We let $\mathbb{S}$ denote the sphere spectrum $\Sigma^\infty S^0$. We can also think about this as $\Sigma^\infty_+ \mathrm{pt}$.

Remark 1.49. The category $\mathrm{Spectra}$ admits internal Homs which has a left adjoint $\otimes_{\mathbb{S}}$. This gives a symmetric monoidal structure to $\mathrm{Spectra}$.

Remark 1.50. For any space X, the constant map $X \to \mathrm{pt}$ induces a map $\Sigma^\infty_+ X \to \mathbb{S}$. On the other hand, for any pointed space X, the canonical map $\mathrm{pt} \to X$ induces a map $\mathbb{S} \to \Sigma^\infty X$. Thus, when X is pointed, we find

$$\Sigma^\infty_+ X = \Sigma^\infty X \oplus \mathbb{S}.$$

1.2.2 Module Categories

We now pass to other rings.

Definition 1.51 ($\mathbb{E}_\infty$-ring)**.** An $\mathbb{E}_\infty$*-ring* R is a spectrum R equipped with a multiplication $R \otimes_{\mathbb{S}} R \to R$ and a unit $\mathbb{S} \to R$, as well as much other data (e.g., many other operations all required to cohere with each other).

One can define modules in an expected way.

Example 1.52. For any $\mathbb{E}_\infty$-ring R, the category $\mathrm{Mod}(R)$ of R-modules continues to be a stable, symmetric monoidal category, where the symmetric monoidal structure is given by some tensor product $\otimes_R$.

Remark 1.53. Conversely, given a symmetric monoidal stable category $\mathcal{C}$, we can recover $\mathcal{C}$ as a module category over the ring $R := \mathrm{End}_{\mathcal{C}}\, 1$, where 1 is a tensor unit. Indeed, if $\mathcal{C} = \mathrm{Mod}_R$, we see that

$$\mathrm{Hom}_{\mathcal{C}}(R, R) = \mathrm{Hom}_{\mathbb{S}}(\mathcal{S}, R)$$

by the tensor–hom adjunction. This is then $\mathrm{Hom}_{\mathbb{S}}(\Sigma^\infty_+ \mathrm{pt}, R)$, which is just $\Omega^\infty R$ because Σ^∞ is left adjoint to Ω^∞.

Here are some constructions with modules.

Definition 1.54. Fix an $\mathbb{E}_\infty$-ring R. For any $x \in \pi_i R$, we receive a spectrum map $x \colon \Sigma^i \mathbb{S} \to R$ and hence an R-module map $\Sigma^i R \to R$. We now define $R\left[x^{-1}\right]$ to be the R-module

$$\mathrm{colim}\left(R \xrightarrow{x} \Sigma^{-i} R \xrightarrow{x} \Sigma^{-2i} R \to \cdots\right).$$

Remark 1.55. By construction, there is a unit $R \to R\left[x^{-1}\right]$. Additionally, by understanding maps from $\mathbb{S}$ to the colimit, we see that $\pi_* R\left[x^{-1}\right] = (\pi_* R)\left[x^{-1}\right]$.

Note that we have only defined $R\left[x^{-1}\right]$ as a module. One may hope to upgrade it to a ring.

Remark 1.56. One can check that the canonical map

$$R \otimes_R R\left[x^{-1}\right] \to R\left[x^{-1}\right] \otimes_R R\left[x^{-1}\right]$$

is an equivalence: on homotopy groups, both sides are $(\pi_* R)\left[x^{-1}\right]$. It follows that $R\left[x^{-1}\right]$ is an idempotent algebra, defined by taking the inverse of the equivalence.

Remark 1.57. Here is another way to find that $R\left[x^{-1}\right]$ is a ring. Let $\mathcal{C}$ be the full subcategory of R-modules M such that the action of x on $\pi_* M$ is invertible. Then $\mathcal{C}$ is closed under the tensor product, so it is a stable symmetric monoidal category, and then Remark 1.53 allows us to find $R\left[x^{-1}\right]$ as the corresponding ring.

1.2.3 Some Free Constructions

Here is a way to construct an $\mathbb{E}_\infty$-ring.

Remark 1.58. Given a spectrum R, note that $X \otimes_{\mathbb{S}} X$ admits a natural action by the symmetric group Σ_2. This produces a functor $\mathrm{B}\Sigma_2 \to \mathrm{Spectra}$ which sends the one object to $X \otimes_{\mathbb{S}} X$ and the nontrivial morphism to the swapping action. We may let $(X \otimes_{\mathbb{S}} X)_{h\Sigma_2}$ denote the "quotient." In general, one can form the "quotient" $(R^{\otimes_{\mathbb{S}} n})_{h\Sigma_n}$.

Remark 1.59. If R is an $\mathbb{E}_\infty$-ring, then there is a canonical map $(R \otimes_{\mathbb{S}} R)_{h\Sigma_2} \to R$ given by the commutativity. The associativity and commutativity produce a map

$$\left(R^{\otimes_{\mathbb{S}} n}\right)_{h\Sigma_n} \to R.$$

Note that the existence of this map requires us to remember the higher coherences present in the definition of the $\mathbb{E}_\infty$-ring R.

Definition 1.60 (free ring)**.** Fix a spectrum X. Then we define the *free $\mathbb{E}_\infty$-ring* $\mathrm{Free}_{\mathbb{E}_\infty} X$ to be

$$\mathrm{Free}_{\mathbb{E}_\infty} X := \mathbb{S} \oplus X \oplus (X \otimes_{\mathbb{S}} X)_{h\Sigma_2} \oplus \cdots.$$

The operation is induced by the construction.

Remark 1.61. The free rings have a grading, but the general ones do not.

Remark 1.62. There is an analogous notion of a free $\mathbb{E}_\infty$-space of a space X, which is

$$\mathrm{Free}_{\mathbb{E}_\infty} X := \mathrm{pt} \sqcup X \sqcup (X \times X)_{h\Sigma_2} \sqcup \cdots.$$

Notably, Σ^∞_+ preserves the symmetric monoidal structure and is a left adjoint, so one knows by pure nonsense that

$$\mathrm{Free}_{\mathbb{E}_\infty} \Sigma^\infty_+ X = \Sigma^\infty_+ \mathrm{Free}_{\mathbb{E}_\infty} X.$$

Recall that $\mathbb{E}_\infty$-spaces are interesting because they allow us to access many spectra.

Remark 1.63. If X is an $\mathbb{E}_\infty$-space, then $\pi_0 X$ is a commutative monoid.

Definition 1.64 (group-like)**.** An $\mathbb{E}_\infty$-space is *group-like* if and only if $\pi_0 X$ is a group.

Remark 1.65. The embedding from group-like $\mathbb{E}_\infty$-spaces to $\mathbb{E}_\infty$-spaces admits a right adjoint $\mathrm{B}\Omega$.

Theorem 1.66 (May)**.** Consider the category of group-like $\mathbb{E}_\infty$-spaces.

(a) There is a fully faithful embedding from the category of group-like $\mathbb{E}_\infty$-spaces to the category of spectra.

(b) Its essential image consists of those spectra with no homotopy groups in negative degrees.

(c) The inverse functor to the fully faithful embedding is Ω^∞.

Note that we currently have two ways to construct an $\mathbb{E}_\infty$-space: there is the free construction, but we could also take $\Omega^\infty \Sigma^\infty_+$ because Ω^∞ automatically outputs $\mathbb{E}_\infty$-spaces. These constructions are related, but a modification is required because Ω^∞ outputs group-like $\mathbb{E}_\infty$-spaces.

Theorem 1.67. For any space X, there is an isomorphism

$$\Omega \mathrm{B}\,\mathrm{Free}_{\mathbb{E}_\infty} X \cong \Omega^\infty \Sigma^\infty_+ X$$

of group-like $\mathbb{E}_\infty$-spaces.

Proof. We use the Yoneda lemma: for any group-like $\mathbb{E}_\infty$-space A, which we may naturally view as a spectrum, we see

$$\begin{aligned}\mathrm{Hom}_{\mathbb{E}_\infty}(\Omega^\infty \Sigma^\infty_+ X, A) &= \mathrm{Hom}_{\mathbb{S}}(\Sigma^\infty_+ X, A)\\ &= \mathrm{Hom}_{\mathrm{Spaces}}(X, A)\\ &= \mathrm{Hom}_{\mathbb{E}_\infty}(\mathrm{Free}_{\mathbb{E}_\infty} X, A)\\ &= \mathrm{Hom}_{\mathbb{E}_\infty}(\Omega \mathrm{B}\,\mathrm{Free}_{\mathbb{E}_\infty} X, A),\end{aligned}$$

where the last equality holds because $\Omega^\infty A$ is group-like. ■

Example 1.68. Note that $\Omega \mathrm{B}\,\mathrm{Free}_{\mathbb{E}_\infty} \mathrm{pt} = \Omega^\infty \Sigma^\infty_+ \mathrm{pt}$, which is $\Omega^\infty \mathbb{S}$. On the other hand, $\mathrm{Free}_{\mathbb{E}_\infty} \mathrm{pt}$ by definition(!) is

$$\mathrm{pt} \sqcup \mathrm{pt} \sqcup (\mathrm{pt} \times \mathrm{pt})_{h\Sigma_2} \sqcup \cdots,$$

which is $\mathrm{B}\Sigma_0 \sqcup \mathrm{B}\Sigma_1 \sqcup \mathrm{B}\Sigma_2 \sqcup \cdots$, also known as the category of finite sets up to isomorphism. Thus, we see that the "group completion" of the category of finite sets (up to isomorphism) is $\Omega^\infty \mathbb{S}$, indicating that the group completion of the category of finite sets ought to be $\mathbb{S}$.

Remark 1.69 (telescope)**.** There is a space sitting between $\mathrm{FinSet}^{\cong}$ and its group completion $\Omega^\infty \mathbb{S}$. It is the colimit of the diagram

$$\mathrm{FinSet}^{\cong} \to \mathrm{FinSet}^{\cong} \to \cdots,$$

where the individual maps "add 1" in the sense that they are $X \mapsto X \sqcup \mathrm{pt}$. As such, this space turns out to be isomorphic to $\mathbb{Z} \times \mathrm{B}\Sigma_\infty$. The induced map $\mathbb{Z} \times \mathrm{B}\Sigma_\infty \to \Omega^\infty \mathbb{S}$ is not an isomorphism, but it becomes an isomorphism after Σ^∞_+.

Remark 1.70. The free $\mathbb{E}_\infty$-ring on $\mathbb{S}$ turns out to be

$$\Sigma^\infty_+ \mathrm{FinSet}^{\cong} = \bigoplus_{n \geq 0} \Sigma^\infty_+ \mathrm{B}\Sigma_n.$$

In particular, we see that taking π_0 gives $\mathbb{Z}[x_0]$. Notably, taking a further group completion will invert by this x_0.

We can also take some free $\mathbb{E}_\infty$-spaces of pointed spaces.

Definition 1.71. Fix a pointed space X. Then $\mathrm{Free}_{\mathbb{E}_\infty,*} X$ is the pushout of the following diagram.

$$\begin{tikzcd} \mathrm{Free}_{\mathbb{E}_\infty} \mathrm{pt} \arrow[r] \arrow[d] & \mathrm{Free}_{\mathbb{E}_\infty} X \arrow[d] \\ \mathrm{pt} \arrow[r] & \mathrm{Free}_{\mathbb{E}_\infty,*} X \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

Theorem 1.72 (May)**.** If X is a pointed space, then

$$\Omega \mathrm{B}\, \mathrm{Free}_{\mathbb{E}_\infty,*} X = \Omega^\infty \Sigma^\infty X.$$

Example 1.73. If X is a connected pointed space, then one can compute that $\mathrm{Free}_{\mathbb{E}_\infty,*} X$ is connected (intuitively, the connected components of $\mathrm{Free}_{\mathbb{E}_\infty} X$ have collapsed). Thus, $\mathrm{Free}_{\mathbb{E}_\infty,*} X$ is already group-like (it's π_0 is just a point!), so $\mathrm{Free}_{\mathbb{E}_\infty,*} X = \Omega^\infty \Sigma^\infty X$. Thus, we may view $\Omega^\infty \Sigma^\infty X$ as the "group completion" of X.

Here is an application of our "free" constructions which does not use the word "free."

Theorem 1.74 (Snaith, Jones)**.** Fix a connected pointed space X. Then

$$\Sigma^\infty_+ \Omega^\infty \Sigma^\infty X = \mathbb{S} \oplus \Sigma^\infty X \oplus (\Sigma^\infty X)^{\otimes 2}_{h\Sigma_2} \oplus \cdots.$$

Proof. Recall that we have a pushout

$$\begin{tikzcd} \mathrm{Free}_{\mathbb{E}_\infty} \mathrm{pt} \arrow[r] \arrow[d] & \mathrm{Free}_{\mathbb{E}_\infty} X \arrow[d] \\ \mathrm{pt} \arrow[r] & \mathrm{Free}_{\mathbb{E}_\infty,*} X \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

of $\mathbb{E}_\infty$-spaces. Note that the bottom-right is $\Omega^\infty \Sigma^\infty X$ by Theorem 1.72. Hitting this with the left adjoint Σ^∞_+ produces a pushout

$$\begin{tikzcd} \mathrm{Free}_{\mathbb{E}_\infty} \mathbb{S} \arrow[r] \arrow[d] & \mathrm{Free}_{\mathbb{E}_\infty} \Sigma^\infty_+ X \arrow[d] \\ \mathbb{S} \arrow[r] & \Sigma^\infty_+ \Omega^\infty \Sigma^\infty X \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

of $\mathbb{E}_\infty$-rings, where the top-right is has $\Sigma^\infty_+ X = \Sigma^\infty X \oplus \mathbb{S}$ and so

$$\begin{aligned} \mathrm{Free}_{\mathbb{E}_\infty} \Sigma^\infty_+ X &= \mathrm{Free}_{\mathbb{E}_\infty} (\Sigma^\infty X \oplus \mathbb{S}) \\ &= \mathrm{Free}_{\mathbb{E}_\infty} \Sigma^\infty X \otimes_{\mathbb{S}} \mathrm{Free}_{\mathbb{E}_\infty} \mathbb{S} \\ &= \mathrm{Free}_{\mathbb{E}_\infty} \Sigma^\infty X. \end{aligned}$$

The result now follows by computing our pushout. ∎

Remark 1.75. This is a remarkable theorem! Basically, it tells us that taking Σ^∞_+ of some group-like $\mathbb{E}_\infty$-space should have homology which splits in a very natural way.

Remark 1.76. It turns out that $\Sigma^\infty_+\Omega^\infty\mathbb{S} = \mathrm{Free}_{\mathbb{E}_\infty}\mathbb{S}\left[x^{-1}\right]$ for some class x arising from the class of a point (in degree one). Note that Theorem 1.74 does not apply here because the space S^0 is not connected! The general statement is that

$$\Sigma^\infty_+\Omega^\infty\Sigma^\infty X = \mathrm{Free}_{\mathbb{E}_\infty}(X)\left[I^{-1}\right],$$

where I is the set of elements in the image of $\pi_0 X \to \pi_0\Sigma^\infty X$. The proof of this statement amounts to "forcing" π_0 to be a group.

1.2.4 The Transfer

Here is a more complicated free example.

1.3 February 9

Here we go.

Remark 1.77. The functor Σ^∞_+ taking $\mathbb{E}_\infty$-spaces to $\mathbb{E}_\infty$-rings admits a left adjoint known as Ω^∞. Note that $\Omega^\infty R$ is merely a (multiplicative) $\mathbb{E}_\infty$-space, which need not be grouplike.

1.3.1 Fun with Σ_2

Let's do a little combinatorics with $\mathrm{FinSet}(\Sigma_2)$.

Example 1.78. There is a single point set, which has automorphism group Σ_2.

Example 1.79. The two-elements sets are $\mathrm{pt}\sqcup\mathrm{pt}$, whose stabilizer is Σ_2, and Σ_2, whose stabilizer is also Σ_2.

Example 1.80. There is a three-element set $\mathrm{pt}^{\sqcup 3}$, whose stabilizer is Σ_3. There is also the three-element set $\mathrm{pt}\sqcup\Sigma_2$, whose stabilizer is Σ_2.

Example 1.81. Consider the four-element set $\Sigma_2\sqcup\Sigma_2$. Then the automorphism group has eight elements (one can swap within the Σ_2s or also swap the two copies of Σ_2), and one can draw a square with each Σ_2 on the diagonals to show that the stabilizer is D_8.

Remark 1.82. There is a fiber sequence

$$\mathrm{B}C_2\times\mathrm{B}C_2 \to \mathrm{B}D_8 \to \mathrm{B}C_2$$

induced by the short exact sequence of groups. It follows that $\mathrm{B}D_8 = (\mathrm{B}C_2\times\mathrm{B}C_2)_{h\Sigma_2}$. In particular, $\mathrm{B}D_8$ lives in $\mathrm{Free}_{\mathbb{E}_\infty}(\mathrm{B}C_2)$.

Example 1.83. Consider the full category of finite Σ_2-sets, considered up to isomorphism. Such a set is a union of some points and some two-element sets with a swapping action. Enumerating all such finite sets, we find that this category is $\mathrm{Free}_{\mathbb{E}_\infty}(\mathrm{pt} \sqcup \mathrm{B}\Sigma_2)$. By Theorem 1.67, we thus see that

$$\Omega \mathrm{B}\, \mathrm{Free}_{\mathbb{E}_\infty}(\mathrm{pt} \sqcup \mathrm{B}\Sigma_2) = \Omega^\infty \left(\mathbb{S} \oplus \Sigma^\infty_+ \mathrm{B}\Sigma_2\right).$$

Here are some linear maps with $\mathrm{FinSet}^{\cong}$.

- There is a forgetful functor $\mathrm{FinSet}(\Sigma_2) \to \mathrm{FinSet}$ which descends to isomorphism classes. It also sends disjoint unions to disjoint unions, so by taking group completion, one receives a map

  $$\Sigma^\infty_+ \mathrm{B}\Sigma_2 \oplus \mathbb{S} \to \mathbb{S},$$

  so there is a "transfer" map $\mathrm{tr}\colon \Sigma^\infty_+ \mathrm{B}\Sigma_2 \to \mathbb{S}$. (Note that we have silently removed our Ω^∞, which we can do because the forgetful functor preserves disjoint unions, so Ω^∞ can be removed by Theorem 1.66 because these things are group-like.) It turns out that this map is given by $(\mathrm{tr}, 1)$.

- There is a map $\mathrm{FinSet} \to \mathrm{FinSet}(\Sigma_2)$ giving a finite set the trivial Σ_2-action. It sends disjoint unions to disjoint unions, so one can argue as above to show that we receive a map

  $$\Omega^\infty \mathbb{S} \to \Omega^\infty \mathbb{S} \times \Omega^\infty \Sigma^\infty_+ \mathrm{B}\Sigma_2$$

  on group completions. It is the inclusion into the first factor, as one can see by tracking around our isomorphisms.

- There is a map $\mathrm{FinSet}(\Sigma_2) \to \mathrm{FinSet}$ which takes the quotient by the Σ_2-action. This time, the corresponding map on group completion

  $$\Sigma^\infty_+ \mathrm{B}\Sigma_2 \oplus \mathbb{S} \to \mathbb{S},$$

  and it takes the form $(\varepsilon, 1)$.

Remark 1.84. It turns out that the map ε is an unstable map, coming from the natural projection $\mathbb{RP}^\infty \to \mathrm{pt}$. The map tr is stable.

Remark 1.85 (Siegel's conjecture)**.** It turns out that all maps $\Sigma^\infty_+ \mathrm{B}\Sigma_2 \to \mathbb{S}$ can be obtained from tr and ε.

1.3.2 The Kahn–Priddy Theorem

Here are some nonlinear maps.

Example 1.86 (squaring)**.** There is a squaring map $x \mapsto x^2$ on $\Omega^\infty \mathbb{S}$. This squares on π_0, so it is not linear. Nonetheless, it sends 0 to 0, so it is at least pointed if 0 is the basepoint. We claim that it vanishes on higher homotopy groups.

Proof. An element in $\pi_i \Omega^\infty \mathbb{S}$ amounts to the data of a map $f\colon S^i \to \Omega^\infty \mathbb{S}$, and we can see that squaring is the map

$$S^i \xrightarrow{\Delta} S^i \times S^i \xrightarrow{(f,f)} \Omega^\infty \mathbb{S} \times \Omega^\infty \mathbb{S} \xrightarrow{m} \Omega^\infty \mathbb{S}.$$

For $i > 0$, it turns out that the map $S^i \to S^i \times S^i$ factors (up to homotopy) with $S^i \vee S^i$, which we can see by pinching some corners. Now, mapping out of $S^i \vee S^i$ is determined by its map on each of the factors, which we can see is nullhomotopic in the composite. It follows that the squaring map $x \mapsto x^2$ is trivial on higher homotopy groups. ∎

Example 1.87 (Norm)**.** There is a canonical factorization

$$\begin{tikzcd} & \Omega^\infty(\Sigma^\infty_+ \mathrm{B}\Sigma_2 \oplus \mathbb{S}) \arrow[d, "(\mathrm{tr},1)"] \\ \Omega^\infty\mathbb{S} \arrow[ur, dashed, "\mathrm{Norm}"] \arrow[r, "x\mapsto x^2"'] & \Omega^\infty\mathbb{S} \end{tikzcd}$$

which defines the norm.

Proof. Combinatorially, the point is that there is a composite

$$\mathrm{FinSet}^{\cong} \to \mathrm{FinSet}(\Sigma_2)^{\cong} \to \mathrm{FinSet}$$

where the left map is squaring, and the right map is forgetful. However, squaring does not preserve disjoint unions, so we cannot just take group completion. However, it turns out that we can move down to the telescope (as in Remark 1.69)

$$\mathbb{Z} \times \mathrm{B}\Sigma_\infty \to \mathrm{Telescope}(\mathrm{FinSet}(\Sigma_2)) \to \mathbb{Z} \times \mathrm{B}\Sigma_\infty,$$

where the middle object is the colimit of $\mathrm{FinSet}(\Sigma_2)$ where we are adding in $\mathrm{pt} \sqcup \mathrm{B}\Sigma_2$ at all levels. We are interested in getting this composite to map to $\Omega^\infty\mathbb{S}$, but by the adjunction, we can instead attempt to get the composite

$$\Sigma^\infty_+(\mathbb{Z} \times \mathrm{B}\Sigma_\infty) \to \Sigma^\infty_+ \mathrm{Telescope}(\mathrm{FinSet}(\Sigma_2)) \to \Sigma^\infty_+(\mathbb{Z} \times \mathrm{B}\Sigma_\infty)$$

to $\mathbb{S}$. However, the above composite is made of group completions—we are using the "Group completion theorem," whose proof is some messing around with the Yoneda lemma. ■

We will want the following result.

Theorem 1.88 (Kahn–Priddy)**.** The map $\pi_* \operatorname{tr}\colon \pi_*\Sigma^\infty_+\mathrm{B}\Sigma_2 \to \pi_*\mathbb{S}$ is surjective in positive degrees.

Proof. Consider the map $\Omega^\infty\mathbb{S} \to \Omega^\infty\Sigma^\infty_+\mathrm{B}\Sigma_2$ by $X \mapsto X^2 \setminus X$, where we are thinking about this as coming from finite sets (mapping to free Σ_2-sets), which upgrades to the group completion (at least as a map to $\Omega^\infty\mathbb{S}$) by the above telescoping trick. The total composite is now $x \mapsto x^2 - x$ on π_0 and so an isomorphism on higher homotopy groups because we showed earlier in Example 1.86 that squaring vanishes on higher homotopy. ■

1.3.3 Nishida Nilpotence

Before we jump into Nishida nilpotence, we will need the D_2 construction.

Definition 1.89 (D_2 construction)**.** Suppose R is an $\mathbb{E}_\infty$-ring, and choose an element $x \in \pi_0 R$, which provides the data of a map $x\colon \mathbb{S} \to R$. We can then form a composite

$$\left(\mathbb{S}^{\otimes 2}\right)_{h\Sigma_2} \to \left(R^{\otimes_{\mathbb{S}} 2}\right)_{h\Sigma_2} \to R,$$

where the left map is $x \otimes x$, and the right map is multiplication. Note that the left object is $\operatorname{colim}_{\mathrm{B}\Sigma_2} \mathbb{S} = \Sigma^\infty_+ \operatorname{colim}_{\mathrm{B}\Sigma_2} \mathrm{pt} = \Sigma^\infty_+\mathrm{B}\Sigma_2$, so we see that we have found an element $D_2(x) \in R^0(\mathbb{RP}^\infty)$.

Example 1.90. Take $R = \mathbb{S}$. Then $D_2(1) = \varepsilon$. Indeed, the multiplication map $\mathbb{S}^{\otimes 2}_{h\Sigma_2} \to \mathbb{S}$ is Σ^∞_+ of the projection $\mathrm{B}\Sigma_2 \to \mathrm{pt}$, so we can see that the total composite is in fact ε.

Example 1.91. Take $R = \mathbb{S}$. To compute $D_2(2)$, we note that

$$(\mathbb{S} \oplus \mathbb{S})^{\otimes 2} = \mathbb{S} \oplus \mathbb{S} \oplus \mathbb{S} \oplus \mathbb{S},$$

and the canonical swapping map on the left fixes two factors and swaps the other two. (Indeed, the isomorphism is basically given by $(x+y)^2 = x^2 + xy + yx + y^2$.) Taking $(-)_{h\Sigma_2}$, it turns out that we get

$$\Sigma^\infty_+ \mathbb{RP}^\infty \to \Sigma^\infty_+ \mathbb{RP}^\infty \oplus \mathbb{S} \oplus \Sigma^\infty_+ \mathbb{RP}^\infty \to \Sigma^\infty_+ \mathbb{RP}^\infty.$$

In particular, $(\mathbb{S} \oplus \mathbb{S})_{h\Sigma_2}$ can be seen to be $\mathbb{S}$ again by considering a left Kan extension of the map $\mathbb{S} \colon \mathrm{pt} \to \mathrm{Spectra}$ to $\mathrm{B}\Sigma_2$. Now, it turns out that the left map is $(1, \mathrm{tr}, 1)$ (one can see that the middle map is transfer by returning to combinatorics of finite sets), so by mapping back to $\mathbb{S}$, we see that $D_2(2) = 2\varepsilon + \mathrm{tr}$.

Remark 1.92. The method of Example 1.91 can show that $D_2(a+b) = D_2(a) + D_2(b) + ab\,\mathrm{tr}$ for any $a, b \in \pi_0\mathbb{S}$. For example, $D_2(4) = r\varepsilon + 6\,\mathrm{tr}$.

And here is our theorem.

Theorem 1.93 (Nishida nilpotence)**.** If $x \in \pi_i\mathbb{S}$ for $i > 0$, then x is nilpotent.

Proof. It suffices to show that $\mathbb{S}\left[x^{-1}\right]$ is the trivial ring, meaning that we want to show that $1 = 0$ in $\pi_0\mathbb{S}\left[x^{-1}\right]$. Now, every higher homotopy group is torsion, so there is at least some positive integer N for which $N = 0$ in $\pi_0\mathbb{S}\left[x^{-1}\right]$. If N admits no prime factors, then we are done. Otherwise, choose a prime p dividing N, and we would like to replace N with N/p, for which it is enough to check that $\mathbb{S}_{(p)}\left[x^{-1}\right] = 0$.

We will give the proof in an example. For example, let's work with $p = 2$ so that we want to show $\mathbb{S}_{(2)}\left[x^{-1}\right] = 0$. By the discussion of the previous paragraph, we may even assume that some power of 2 vanishes. Further specializing with loss of generality, let's say that 4 vanishes. (The same idea turns out to work in general.) This means that the composite

$$\mathbb{S} \xrightarrow{4} \mathbb{S} \to \mathbb{S}_{(2)}\left[x^{-1}\right]$$

is nullhomotopic. Squaring, we can form the diagram

$$\begin{tikzcd}
\mathbb{S}^{\otimes 2}_{h\Sigma_2} \arrow[r, "4"] & \mathbb{S}^{\otimes 2}_{h\Sigma_2} \arrow[r] \arrow[d, equal] & \mathbb{S}_{(2)}\left[x^{-1}\right]^{\otimes 2} \arrow[d, "m"] \\
& \mathbb{S} \arrow[r] & \mathbb{S}_{(2)}\left[x^{-1}\right]
\end{tikzcd}$$

where the top composite is nullhomotopic. But the zigzag path is $D_2(4)$, so we conclude that the composite

$$\mathbb{S}^{\otimes 2}_{h\Sigma_2} \xrightarrow{D_2(4)} \mathbb{S} \to \mathbb{S}_{(2)}\left[x^{-1}\right]$$

is nullhomotopic. Now, $D_2(4) = 4 + 6\,\mathrm{tr}$, so because $4 = 0$, we further see that the composite

$$\mathbb{S}^{\otimes 2}_{h\Sigma_2} \xrightarrow{2\,\mathrm{tr}} \mathbb{S} \to \mathbb{S}_{(2)}\left[x^{-1}\right]$$

is nullhomotopic. But now Theorem 1.88 grants us a map $\mathbb{S}^i \to \Sigma^\infty_+ \mathbb{RP}^\infty$ which goes back to x under $\pi_*\,\mathrm{tr}$, so it follows that

$$\mathbb{S}^i \xrightarrow{2x} \mathbb{S} \to \mathbb{S}_{(2)}\left[x^{-1}\right]$$

is nullhomotopic. Thus, $2x = 0$ in $\mathbb{S}_{(2)}\left[x^{-1}\right]$, so $2 = 0$.

Let's at least say a sentence or two about why this should work in general. Continuing with $p = 2$, the point is that $D_2(2a) = 2D_2(a) + a^2\,\mathrm{tr}$, and it turns out that the trace term again has lower 2-adic valuation. Thus, one can expect to be able to induct and handle general powers of 2. For other primes p, we need a D_p construction, and again, the point is that the multinomial coefficients in an expansion $(a_1 + \cdots + a_p)^p$ will have smaller p-adic valuation. ∎

Remark 1.94. There is a specific element in $\kappa \in \pi_{24}\mathbb{S}$ for which we do not know if $\kappa^{11} = 0$ or if merely $\kappa^{12} = 0$.

Remark 1.95. One can upgrade this argument to show the following: for an $\mathbb{E}_\infty$-ring R, one has $R = 0$ if and only if $\mathbb{Z} \otimes_{\mathbb{S}} R = 0$. (This truly uses the $\mathbb{E}_\infty$-structure: it is false if one tries to work with $\mathbb{E}_n$-rings for any finite n!) In particular, plugging in $\mathbb{S}\left[x^{-1}\right]$ for R recovers Theorem 1.93.

Theme 2
Formal Groups and Slopes

2.1 February 11

Today, we say a little bit about MU. There are many existing sources: there is Adams's red book, Ravenel's green book, Wilson's *An Introduction and Sampler*, Lurie's chromatic homotopy notes, Pstragowki's book, and Peterson's book.

2.1.1 Formal Group Laws

Let's start with some pure algebra.

Definition 2.1 (formal group law)**.** Fix a commutative ring R. Then a *formal group law* over R is a power series $f \in R[[x, y]]$ such that

(a) $f(x, 0) = f(0, x) = x$,

(b) $f(x, y) = f(y, x)$, and

(c) $f(f(x, y), z) = f(x, f(y, z))$.

We may write $x +_f y$ for $f(x, y)$.

Remark 2.2. One is supposed to think of f as providing the formula for an actual group law, under the assumption that f always converges. More rigorously, these are supposed to be additions on $\widehat{\mathbb{A}}^1$.

Definition 2.3 (isomorphism)**.** Fix a commutative ring R. An *isomorphism* $g\colon f_1 \to f_2$ of formal group laws is a power series $g \in R[[x]]$ for which $g(0) = 0$, $g'(0) \in R^\times$ and

$$g(x +_{f_1} y) = x +_{f_2} y.$$

We say that g is *strict* if and only if $g'(0) = 1$.

Example 2.4. There is the additive formal group law $f(x, y) := x + y$.

Example 2.5. There is the multiplicative formal group law $f(x, y) = x + y + xy$.

There is an important "universal" formal group law.

Definition 2.6 (Lazard ring)**.** The *Lazard ring* L is defined to co-represent the functor sending a ring R of all formal group laws over R. Explicitly, we can realize it as a quotient of the ring $\mathbb{Z}[\{a_{ij}\}_{ij}]$ by the relations required to make the power series

$$\sum_{i,j\geq 0} a_{ij}x^i y^j$$

into a formal group law. We give L a grading by $\deg a_{ij} := 2(i+j-1)$.

Definition 2.7. We define $\mathbb{Z}[b_1, b_2, \ldots]$ to be the universal ring equipped with a formal group law and an isomorphism of this formal group law. Explicitly, it is a ring $\mathbb{Z}[b_1, b_2, \ldots]$ equipped with a map $\varphi\colon L \to \mathbb{Z}[b_1, b_2, \ldots]$ to classify $g\left(g^{-1}(x) + g^{-1}(y)\right)$, where $g(x) := x + b_1x^2 + b_2x^3$. We give this ring a grading by $\deg b_i = 2i$, so the map φ is graded.

Remark 2.8. One can see that $L[b_1, b_2, \ldots]$ classifies two formal group laws and a choice of strict isomorphism between them. Indeed, L provides the formal group law, and the $b_\bullet$s provide the strict isomorphism.

Theorem 2.9 (Lazard)**.** For all $n > 0$, the map $\varphi_{2n}\colon L_{2n} \to \mathbb{Z}[b_1, b_2, \ldots]_{2n}$ is an injection after taking a quotient by the decomposable elements. Explicitly, the target is isomorphic to $\mathbb{Z}b_{2n}$, and the image is full if $n+1$ is not a prime power; otherwise, the image is $p\mathbb{Z}$ if $n+1$ is a power of p.

Idea. The hard part is to show that L_{2n} modulo decomposable elements is isomorphic to $\mathbb{Z}$. Accordingly, let this quotient be Q_{2n}. Then one needs to exhibit a natural isomorphism $\mathrm{Hom}_{\mathbb{Z}}(Q_{2n}, M) \cong M$, but one finds that maps $Q_{2n} \to M$ are given by formal group laws over the square zero extension $\mathbb{Z} \oplus M$ of the form

$$x + y + \sum_{i+j=n+1} x^i y^j.$$

Then one needs to classify such things, which is an algebra problem with binomial coefficients. ■

Corollary 2.10. For each i, choose $x_i \in L$ of degree $2i$ which generates L_{2i} modulo decomposable elements. Then the induced map

$$\mathbb{Z}[x_1, x_2, \ldots] \to L$$

is an isomorphism of rings.

Proof. This follows from Theorem 2.9: indeed, Theorem 2.9 shows that we have produced an isomorphism after passing to the associated graded ring for the relevant filtration, which is enough to show that we have an isomorphism. ■

Thus, L is just a polynomial ring! However, this presentation is not very helpful because it requires infinitely many choices. Alternatively, the map φ is an isomorphism after taking tensor product with $\mathbb{Q}$ by Theorem 2.9, but taking this tensor product may lose information we care about.

2.1.2 The Spectrum MU

We now return to algebraic topology. We would like to make sense of the following theorem.

Theorem 2.11 (Quillen)**.** There is an $\mathbb{E}_\infty$-ring MU satisfying the following.

(a) $\pi_*\mathrm{MU} \cong L$.

(b) The induced map $\pi_*\mathrm{MU} \to \pi_*(\mathbb{Z} \otimes_{\mathbb{S}} \mathrm{MU})$ is isomorphic to φ.

(c) There are isomorphisms $\pi_*(\mathrm{MU} \otimes_{\mathbb{S}} \mathrm{MU}) \cong \pi_*\mathrm{MU}[b_1, b_2, \ldots] \cong L[b_1, b_2, \ldots]$ such that the two induced maps $\pi_*\mathrm{MU} \to \pi_*(\mathrm{MU} \otimes_{\mathbb{S}} \mathrm{MU})$ correspond to the source and target of the universal formal group laws of $L[b_1, b_2, \ldots]$.

Of course, we will have to start by constructing MU.

Remark 2.12. There is a functor $S\colon \mathrm{Vec}_{\mathbb{C}} \to \mathrm{Top}_*$ of ordinary category given by taking the one-point compactification. It satisfies $S(V \oplus W) = SV \wedge SW$, so S is symmetric monoidal, and we get a map

$$\bigsqcup_{n \geq 0} \mathrm{BU}(n) \to \mathrm{Top}_* \to \mathrm{Spaces}_*$$

sending V to $\Sigma^\infty S^V$.

Definition 2.13 (Picard group)**.** Fix an $\mathbb{E}_\infty$-ring R. Then $\mathrm{Pic}\, R$ is the full subcategory of $\mathrm{Mod}(R)$ of $\otimes$-invertible objects, only equipped with the isomorphisms.

Example 2.14. The only invertible objects in $\mathrm{Pic}\,\mathbb{S}$ are powers of $\mathbb{S}$. Here is a sketch: any invertible object remains invertible after $- \otimes_{\mathbb{S}} \mathbb{Z}$, so it becomes an invertible object in $\mathrm{Mod}(\mathbb{Z})$. But the only invertible chain complexes are shifts of $\mathbb{Z}$, so we know that any invertible object has homology supported in one degree. By the Hurewicz theorem, it therefore has homotopy starting in one degree (and zero below), so we receive a map from a sphere, and one can check that it is an isomorphism by a variant of Whitehead's theorem.

Remark 2.15. Note that $\mathrm{Pic}\, R$ is in fact a group-like $\mathbb{E}_\infty$-space, where the operation is given by $\otimes$.

Remark 2.16. It thus turns out that the map $\bigsqcup_{n \geq 0} \mathrm{BU}(n) \to \mathrm{Spectra}$ given by $V \mapsto \Sigma^\infty SV$ factors through $\mathrm{Pic}\,\mathbb{S}$ because it only produces powers of spheres.

Definition 2.17 (Thom spectrum)**.** Fix an $\mathbb{E}_\infty$-ring R. If $f\colon X \to \mathrm{Pic}\, R$ is a map of groupoids, then we define the *Thom spectrum* $\mathrm{Thom}(f)$ to be the colimit of the diagram

$$X \to \mathrm{Pic}\, R \to \mathrm{Mod}(R).$$

Remark 2.18. Intuitively, the map $f\colon X \to \mathrm{Pic}\, R$ is a bundle of invertible spectra on R, so we are recovering a classical notion.

Example 2.19. We define the map $\bigsqcup_{n \geq 0} \mathrm{BU}(n) \to \mathrm{Pic}\,\mathbb{S}$ to have Thom spectrum $\bigoplus_{n \geq 0} \mathrm{MU}(n)$.

Example 2.20. We claim that $\mathrm{MU}(1)$ is $\Sigma^\infty \mathbb{CP}^\infty$.

Proof. We are looking at the colimit which $\mathrm{BU}(1) \to \mathrm{Spectra}$ which sends a line bundle $\mathcal{L} \in \mathrm{B}S^1$ to the spectrum $\Sigma^\infty S\mathcal{L}$. Let $S^1\mathcal{L}$ be the unit sphere in $\mathcal{L}$, and then there is a homotopy fiber sequence

$$S^1\mathcal{L}_+ \to \mathrm{pt}_+ \to S\mathcal{L}.$$

Intuitively, this homotopy fiber sequence is gluing some part of $S\mathcal{L}$ against a point to produce a sphere with a point.

It follows that $\mathrm{MU}(1)$ can be computed as

$$\begin{aligned}\operatorname*{colim}_{\mathrm{B}S^1} \Sigma^\infty S\mathcal{L} &= \Sigma^\infty \operatorname*{colim}_{\mathrm{B}S^1} S\mathcal{L} \\ &= \Sigma^\infty \operatorname*{colim}_{\mathrm{B}S^1} \operatorname{cofiber}\left(S^1\mathcal{L}_+ \to \mathrm{pt}_+\right) \\ &= \operatorname{cofiber}\left(\operatorname*{colim}_{\mathrm{B}S^1} \Sigma^\infty_+ S^1\mathcal{L} \to \operatorname*{colim}_{\mathrm{B}S^1} \Sigma^\infty_+ \mathrm{pt}\right).\end{aligned}$$

Now, the right colimit has Σ^∞_+ come out, so it is $\Sigma^\infty_+ \mathbb{CP}^\infty$. The left colimit is the colimit of the group S^1 acting on itself, which is the sphere $\mathbb{S}$. Now, the map $\mathbb{S} \to \Sigma^\infty_+ \mathbb{CP}^\infty$ just gets rid of the point, so the cofiber is $\Sigma^\infty \mathbb{CP}^\infty$. ■

Now, we recognize that we have a map $\bigsqcup_{n\geq 0} \mathrm{BU}(n) \to \operatorname{Pic}\mathbb{S}$ which has target equal to a grouplike $\mathbb{E}_\infty$-space. Thus, we can take a group completion to produce a map

$$\Omega\mathrm{B}\bigsqcup_{n\geq 0} \mathrm{BU}(n) \to \operatorname{Pic}\mathbb{S}.$$

The left-hand side is just $\mathrm{BU} \times \mathbb{Z}$ (here, BU is the colimit BU_∞), and it has no negative homotopy groups, so we can realize it as some $\Omega^\infty \mathrm{ku}$ for some ku.

Definition 2.21. Fix everything as above. Then MU is the Thom spectrum of the sequence

$$\mathrm{BU} \to \mathrm{BU} \times \mathbb{Z} \to \operatorname{Pic}\mathbb{S}.$$

Remark 2.22. One can realize the left map as Ω^∞ applied to the sequence $\tau_{\geq 2}\mathrm{ku} \to \mathrm{ku} \to \mathrm{pic}\mathbb{S}$, where $\mathrm{pic}\,\mathbb{S}$ is the space for which $\Omega^\infty \mathrm{pic}\,\mathbb{S} = \operatorname{Pic}\mathbb{S}$.

Remark 2.23. It was important to mention Pic so that we could take group completion.

Remark 2.24. Our definition of MU automatically gives it the structure of a group-like $\mathbb{E}_\infty$-space. It is already a colimit valued in $\mathbb{S}$-modules, so MU becomes an $\mathbb{E}_\infty$-ring!

Remark 2.25. The Thom spectrum of the map

$$\Omega\mathrm{B}\bigsqcup_{n\geq 0} \mathrm{BU}(n) \to \operatorname{Pic}\mathbb{S}$$

produces a space MUP, which is isomorphic to $\bigoplus_{k\in\mathbb{Z}} \Sigma^k \mathrm{MU}$. Its ring structure is not unique! It is an open question if the ring structure on MU is unique.

2.1.3 Complex Orientations

Let's find some structure on MU.

Remark 2.26. By taking Thom spectra of the map

$$\mathbb{CP}^\infty \to \bigsqcup_{n\geq 0} \mathrm{BU}(n) \to \mathrm{BU} \times \mathbb{Z} \to \operatorname{Pic} \mathbb{S}$$

produces a map $\mathrm{MU}(1) \to \mathrm{MUP}$. Note that the copy of $\mathbb{Z}$ here keeps track of the dimension of the ambient vector bundle, so our copy of $\mathrm{BU}(1)$ will land in the first component of $\mathrm{BU} \times \mathbb{Z}$. Expanding out MUP, we see that we have induced a map $\Sigma^\infty \mathbb{CP}^\infty \to \Sigma^2 \mathrm{MU}$.

The sort of gadget in Remark 2.26 is important enough to deserve a name.

Definition 2.27 (complex orientation)**.** Fix an $\mathbb{E}_\infty$-ring R. A *complex orientation* is a map $\Sigma^\infty \mathbb{CP}^\infty \to \Sigma^2 R$ for which the composite

$$\Sigma^\infty \mathbb{CP}^1 \to \Sigma^\infty \mathbb{CP}^\infty \to \Sigma^2 R$$

is the unit map $\Sigma^2 \mathbb{S} \to \Sigma^2 R$.

Complex orientations give rise to formal group laws. Let's explain how this is done.

Remark 2.28. A choice of complex orientation yields an isomorphism

$$R^* \mathbb{CP}^\infty \cong \pi_{-*} R[[t]],$$

where $t \in R^2 \mathbb{CP}^\infty$ is the class induced by the complex orientation. Indeed, this is some spectral sequence calculation. The point is that the Atiyah–Hirzebruch spectral sequence tells us that $R^* \mathbb{CP}^\infty$ certainly looks like $\pi_{-*} R[[t]]$ on E_2, and there cannot be differentials making this smaller because t is a genuine cohomology class.

Remark 2.29. One can similarly show that a complex orientation on R produces an isomorphism

$$R^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty) \cong \pi_{-*} R[[x, y]].$$

Remark 2.30. The map $\mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ given by the tensor product of lines is the same data as a map $\Sigma^2 \mathbb{Z} \oplus \Sigma^2 \mathbb{Z} \to \Sigma^2 \mathbb{Z}$ after Ω^∞. This then induces a map $\pi_{-*} R[[t]] \to \pi_{-*} R[[x, y]]$. The image of t turns out to be a formal group law, which one can see all the back from the line bundle maps.

Thus, we have a formal group law on $\pi_* \mathrm{MU}$, so we receive a map $L \to \pi_* \mathrm{MU}$. Theorem 2.11 will tell us that this map is an isomorphism. One next proceeds by producing maps

$$\pi_* \mathrm{MU} \to \mathrm{H}_*(\mathrm{MU}; \mathbb{Z}) \to \mathrm{H}_*(\mathrm{MU}; \mathbb{Q}).$$

Thus, it is profitable to understand the target rings. To this end, note that we have a square

$$\begin{tikzcd} \operatorname{Pic} \mathbb{S} \arrow[r] \arrow[d] & \mathrm{Mod}(\mathbb{S}) \arrow[d, "-\otimes \mathbb{Z}"] \\ \operatorname{Pic} \mathbb{Z} \arrow[r] & \mathrm{Mod}(\mathbb{Z}) \end{tikzcd}$$

where the vertical maps are given by base-change, and the horizontal maps are inclusions. Observe that $\pi_0 \operatorname{Pic} \mathbb{Z} = \mathbb{Z}$ because the invertible chain complexes are all just $\mathbb{Z}$ shifted by some degree. It then turns out that $\tau_{\geq 1} \operatorname{Pic} \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ embeds into homomorphisms $\mathbb{Z} \to \mathbb{Z}$, which is $\Omega^\infty \mathbb{Z}$.

Now, to compute $\mathrm{H}_*(\mathrm{MU};\mathbb{Z}) = \pi_*(\mathrm{MU} \otimes_{\mathbb{S}} \mathbb{Z})$, we are interested in making BU map all the way to $\mathrm{Mod}(\mathbb{Z})$. But $\mathrm{Pic}\,\mathbb{Z}$ only has homotopy groups below 1, and BU is simply connected, so the induced map $\mathrm{BU} \to \mathrm{Mod}(\mathbb{Z})$ is nullhomotopic. It follows that

$$\mathrm{MU} \otimes_{\mathbb{S}} \mathbb{Z} = \Sigma^\infty_+ \mathrm{BU} \otimes_{\mathbb{S}} \mathbb{Z},$$

so $\pi_*(\mathrm{MU} \otimes_{\mathbb{S}} \mathbb{Z}) = \mathrm{H}_*(\mathrm{BU};\mathbb{Z})$. Then one can make some arguments with spectral sequences and so on.

2.2 February 17

Today we continue.

2.2.1 Maps from MU

Let's try to better understand MU. For example, we can try to understand its functor of points.

Remark 2.31. Choose an $\mathbb{E}_\infty$-ring R. If the composite

$$\mathrm{BU} \to \mathrm{Pic}\,\mathbb{S} \to \mathrm{Pic}\,R$$

is nullhomotopic, then taking the colimit to $\mathrm{Mod}(R)$ shows that $R \otimes_{\mathbb{S}} \mathrm{MU}$ is a constant colimit

$$\operatorname*{colim}_{\mathrm{BU}} R = R \otimes \operatorname*{colim}_{\mathrm{BU}} \mathbb{S} = R \otimes \Sigma^\infty_+ \mathrm{BU}.$$

But note BU is already pointed, so $\Sigma^\infty_+ \mathrm{BU} = \Sigma^\infty \mathrm{BU} \oplus \mathbb{S}$, so we receive an R-module map $R \otimes_{\mathbb{S}} \mathrm{MU} \to R$, which is the same data as an $\mathbb{S}$-module map $\mathrm{MU} \to R$.

Remark 2.32. This discussion of nullhomotopies has a precursor: the data of a complex orientation on E is equivalent to the data of a nullhomotopy of the composite $\mathbb{CP}^\infty \to \mathrm{Pic}\,\mathbb{S} \to \mathrm{Pic}\,E$. Indeed, simply apply the adjunction between Σ^2 and Ω^2.

Remark 2.33. One can upgrade Remark 2.31 to show that the space of unital maps $\mathrm{MU} \to R$ is equivalent to the data of a nullhomotopy of the composite

$$\mathrm{BU} \to \mathrm{Pic}\,\mathbb{S} \to \mathrm{Pic}\,R$$

in Spaces_*. (Indeed, providing a map out of MU is a map out of a certain colimit, and unwinding the data finds the required nullhomotopy.) Furthermore, the space of $\mathbb{E}_n$-ring maps $\mathrm{MU} \to R$ is equivalent to the space of nullhomotopies of the same composite in the category of n-fold loop spaces.

We can work out the above remark in some controlled situations. Here are the "$\mathbb{Z}$-points" of MU.

Example 2.34. The map $\mathrm{BU} \to \mathrm{Pic}\,\mathbb{S} \to \mathrm{Pic}\,\mathbb{Z}$ is nullhomotopic in the category of infinite loop spaces because $\mathrm{Pic}\,\mathbb{Z}$ only has homotopy in two degrees (namely 0 and 1), and BU has no homology there. Thus, there is an isomorphism $\mathbb{Z} \otimes_{\mathbb{S}} \Sigma^\infty_+ \mathrm{BU} \cong \mathbb{Z} \otimes_{\mathbb{S}} \mathrm{MU}$ of $\mathbb{E}_\infty$-rings, so one can compute

$$\mathrm{H}_*(\mathrm{MU};\mathbb{Z}) \cong \mathrm{H}_*(\mathrm{BU};\mathbb{Z}),$$

and then one can calculate $\mathrm{H}^*(\mathrm{BU};\mathbb{Z}) \cong \mathbb{Z}[b_1, b_2, \ldots]$.

Remark 2.35. Let's explain where the b_is may come from. The natural map $\mathrm{MU}(1) \to \Sigma^2\mathrm{MU}$ (placing $\mathrm{MU}(1)$ in the correct degree) is the same as the complex orientation $\Sigma^\infty\mathbb{CP}^\infty \to \Sigma^2\mathrm{MU}$. This then induces a map

$$\mathbb{Z} \otimes_{\mathbb{S}} \Sigma^\infty\mathbb{CP}^\infty \to \mathbb{Z} \otimes_{\mathbb{S}} \Sigma^2\mathrm{MU}.$$

But now the homology of $\mathbb{CP}^\infty$ is the free $\mathbb{Z}$-module on the generators $b_1, b_2, \ldots$.

Here are the MU-points of MU.

Example 2.36. There is a canonical identity $\mathrm{MU} \to \mathrm{MU}$, which thus induces a nullhomotopy $\mathrm{BU} \to \operatorname{Pic}\mathbb{S} \to \operatorname{Pic}\mathrm{MU}$ in the category of infinite loop spaces. Thus, we see that

$$\mathrm{MU}_*\mathrm{MU} = \mathrm{MU}_*\Sigma^\infty_+\mathrm{BU}.$$

Now, the Atiyah–Hirzebruch spectral sequence degenerates, so this is $\pi_*\mathrm{MU}[b_1, b_2, \ldots]$. One way to see it degenerates is to know that $\pi_*\mathrm{MU}$ is concentrated in even degrees (via Theorem 2.11). There is also an easier way to see this, basically by directly showing that the b_is are permanent cycles, which by the Leibniz rule is equivalent to showing that b_1 is permanent, which can be checked more directly from $\mathrm{MU}(1)$.

The moral of the previous example is that $\mathrm{MU}_*\mathrm{MU}$ is the universal ring for a pair of formal group laws and a choice of strict isomorphism between them.

Proposition 2.37. Fix an even $\mathbb{E}_\infty$-ring E. Then a complex orientation of E extends to an $\mathbb{E}_2$-map $\mathrm{MU} \to E$.

Proof. Note that there is a canonical line bundle $\mathcal{L}\colon \mathbb{CP}^\infty \to \mathrm{BU} \times \mathbb{Z}$, so there is a map $(\mathcal{L} - 1)\colon \mathbb{CP}^\infty \to \mathrm{BU} \times \mathbb{Z}$, which then maps to $\operatorname{Pic}\mathbb{S}$ as before. As discussed last class, we see the Thom spectrum of $(\mathcal{L} - 1)$ is $\Sigma^{-2}\mathrm{MU}(1)$. The moral is that we have some composite

$$\mathbb{CP}^\infty \to \mathrm{BU} \to \operatorname{Pic}\mathbb{S} \to \operatorname{Pic} E.$$

Taking Σ^2 yields

$$\Sigma^2\mathbb{CP}^\infty \to \mathrm{BSU} \to \mathrm{B}^2 \operatorname{Pic}\mathbb{S} \to \mathrm{B}^2 \operatorname{Pic} E.$$

A nullhomotopy of the full sequence is a complex orientation on E, but a choice of nullhomotopy $\mathrm{BSU} \to \mathrm{B}^2 \operatorname{Pic} E$ on its own is an $\mathbb{E}_2$-map $\mathrm{MU} \to E$. Thus, we want to show that the former maps always extend to the latter maps. By splitting up $\mathrm{B}^2 \operatorname{Pic} E = \Omega^\infty\Sigma^2 \operatorname{pic} E$ into its Postnikov tower, it is enough to check that the maps

$$\mathrm{H}^*(\mathrm{BSU}; -) \to \mathrm{H}^*\left(\Sigma^2\mathbb{CP}^\infty; -\right)$$

are surjective, which can be checked by calculations with the spectral sequence. ■

Remark 2.38. As a consequence, we note that an even $\mathbb{E}_\infty$-ring E makes $E^*\mathbb{CP}^\infty$ into a formal group. Choosing coordinates means choosing an isomorphism $E^*\mathbb{CP}^\infty \cong \pi_{-*}E[[t]]$, which means choosing a formal group law up to isomorphism and thus induces a map $L \to \pi_*E$. Our discussion is intended to motivate the fact that choosing a coordinate alternatively should come from a choice of complex orientation and therefore the map $L \to \pi_*E$ comes from an $\mathbb{E}_2$-ring map $\mathrm{MU} \to E$. It is an interesting (open) question when one can upgrade this $\mathbb{E}_2$-ring map into an $\mathbb{E}_\infty$-ring map.

2.2.2 Localization

Let's try to apply some of this discussion.

Notation 2.39. Given a formal group law f, we define the power series $[n]_f(x)$ as $x +_f \cdots +_f x$, where x is added to itself n times. We will write $[n]$ for $[n]_f$ if there is no possibility of confusion.

Example 2.40. In the Lazard ring $L_{(p)}$, one finds that

$$[3](x) = 3x - 8x_2x^3 + 72x_2^2x^5 - 840x_2^3x^7 + \cdots ,$$

where the $x_\bullet$s are the Hazewinkel generators of $L_{(p)} \cong \mathbb{Z}[x_1, x_2, x_3, \ldots]$. (Here, x_i has degree $2i$, and x has degree -2, so the above polynomial is homogeneous.)

Remark 2.41. With the Hazewinkel generators, one can show that the universal formal group law f over $L_{(p)}$ is isomorphic to one that only involves the coefficients $x_{p^\bullet-1}$, and in fact, up to units, x_{p^i-1} is the coefficient of x^{p^i}. Thus, setting $v_i := x^{p^i-1}$, we see that the ring

$$\mathbb{Z}_{(p)}[v_1, v_2, \ldots]$$

carries the universal formal p-typical group law up isomorphism! (Here, p-typical means that our coefficients x_j vanish for j which is not one less than a power of p.) The point is that we only have to care about these coefficients v_i to define such a formal group law.

Example 2.42. There is an idempotent ring endomorphism $f\colon L_{(p)} \to L_{(p)}$ sending x_i to itself if i is one less than a power of p and sending x_i to zero otherwise. Note that the colimit of the diagram

$$L_{(p)} \xrightarrow{f} L_{(p)} \xrightarrow{f} \cdots$$

is just $\mathbb{Z}_{(p)}[v_1, v_2, \ldots]$.

The above remarks prove the following.

Proposition 2.43. There is an $\mathbb{E}_2$-ring map $\mathrm{MU}_{(p)} \to \mathrm{MU}_{(p)}$ whose telescope (in the category of $\mathbb{E}_2$-rings) is an $\mathbb{E}_2$-ring BP with

$$\pi_*\mathrm{BP} \cong \mathbb{Z}_{(p)}[v_1, v_2, \ldots],$$

where v_i lives in degree $2p^i - 2$.

Proof. Let's explain how to produce the map $\mathrm{MU}_{(p)} \to \mathrm{MU}_{(p)}$. Indeed, the algebraic map $L \to \pi_* \mathrm{MU}_{(p)}$ (given by the change of coordinates just described) lifts to an $\mathbb{E}_2$-map $\mathrm{MU} \to \mathrm{MU}_{(p)}$ (because this is how one maps out of MU), which is then further localized. ■

The moral of BP is that we have removed all of our "junk" generators.

Remark 2.44. It is known that BP is not $\mathbb{E}_\infty$, but it is $\mathbb{E}_4$. Exactly the highest $\mathbb{E}_n$ which works is currently open.

Remark 2.45. As before, one can compute $\mathrm{BP}_*\mathrm{BP}$ as $\mathbb{Z}_{(p)}[v_1, v_2, \ldots][t_1, t_2, \ldots]$, which classifies two formal p-local formal group laws and a choice of p-typical strict isomorphism between them. Namely, the isomorphism is given by $x \mapsto x +_f t_1x^p +_f t_2x^{p^2} +_f \cdots$.

2.3 February 18

Let's start using MU to understand $\mathbb{S}$.

2.3.1 Filtrations for Fun and Profit

Here is how we will use MU to get to $\mathbb{S}$.

Lemma 2.46. The spectrum $\mathbb{S}$ is the limit of the following diagram.

$$\begin{tikzcd} \mathrm{MU} \arrow[r, shift left] \arrow[r, shift right] & \mathrm{MU}^{\otimes 2} \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \mathrm{MU}^{\otimes 3} \arrow[r, shift left=3] \arrow[r, shift right=3] & \cdots \end{tikzcd}$$

Proof. This follows from the following general fact: if $A \to B$ is a map of $\mathbb{E}_\infty$-rings whose fiber has homotopy groups only in positive degree, then A is the limit of the Čech nerve of B over A (which is the given colimit). This fact is checked by considering the partial limits and showing that they induce isomorphisms on higher and higher homotopy groups, so we get an isomorphism in the limit. Anyway, it is enough to check that $\mathbb{S} \to \mathrm{MU}$ is an isomorphism in degree zero, which is true by Theorem 2.11. ■

Notation 2.47. We let $\mathrm{fil}^n \mathbb{S}$ be the limit of the following diagram.

$$\begin{tikzcd} \tau_{\geq 2n}\mathrm{MU} \arrow[r, shift left] \arrow[r, shift right] & \tau_{\geq 2n}\mathrm{MU}^{\otimes 2} \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \tau_{\geq 2n}\mathrm{MU}^{\otimes 3} \arrow[r, shift left=3] \arrow[r, shift right=3] & \cdots \end{tikzcd}$$

Using $2n$ instead of n is a convention, motivated by Theorem 2.11.

Example 2.48. Note that MU has only homotopy groups in nonnegative degrees, we see that $\mathrm{fil}^n \mathbb{S} = \mathbb{S}$ for $n \leq 0$. It follows that the colimit of the diagram

$$\cdots \to \mathrm{fil}^3 \mathbb{S} \to \mathrm{fil}^2 \mathbb{S} \to \mathrm{fil}^1 \mathbb{S} \to \mathrm{fil}^0 \mathbb{S}$$

is $\mathbb{S}$. In fact, the limit of this diagram is 0, which more or less follows from Lemma 2.46.

We are thus motivated to say something about filtrations of spectra.

Definition 2.49. A *filtered spectrum* X_* is a diagram of the form

$$\cdots \to X_{-2} \to X_{-1} \to X_0 \to X_1 \to X_2 \to \cdots.$$

A morphism of such filtered spectra is a morphism of the underlying diagrams, including homotopies for all the relevant squares.

Remark 2.50. It turns out that the category of filtered spectra is then a stable ∞-category. It is further symmetric monoidal, with $\otimes$ given by

$$(X_* \otimes Y_*)_n := \operatorname*{colim}_{i+j \geq n} (X_i \otimes Y_j).$$

For example, if the filtered spectrum is in fact given from a "graded spectrum" (so that the internal maps between the X_*s and Y_*s are all split embeddings), then the right-hand side is $\bigoplus_{i+j=n}(X_i \otimes Y_j)$. This formula shows that the object $\cdots 0 \to 0 \to \mathbb{S} \to \mathbb{S} \to \cdots$ is the identity.

Example 2.51. If E is an $\mathbb{E}_\infty$-ring, then $\{\tau_{\geq 2*}E\}_n$ is an $\mathbb{E}_\infty$-algebra valued in filtered complexes.

Example 2.52. It now makes sense to say that $\mathrm{fil}^* \mathbb{S}$ is the limit of the Čech nerve

$$\tau_{\geq 2*}\mathrm{MU} \to \tau_{\geq 2*}\mathrm{MU}^{\otimes 2} \to \cdots$$

in the category of filtered $\mathbb{E}_\infty$-rings. It follows that $\mathrm{fil}^* \mathbb{S}$ gains the structure of an $\mathbb{E}_\infty$-algebra valued in the category of filtered spectra.

The benefit of doing category theory is that we can make categories.

Definition 2.53 (synthetic spectrum)**.** A *synthetic spectrum* is a module of $\mathrm{fil}^* \mathbb{S}$ in the category of filtered spectra. The (stable ∞-)category of synthetic spectra is denoted $\mathrm{SynSpectra}$.

Example 2.54. There is a functor ν which takes a spectrum X and produces the synthetic spectrum

$$\lim \tau_{\geq 2*}\left(X \otimes \mathrm{MU}^{\otimes \bullet + 1}\right).$$

Here is the point of working with synthetic spectra.

Proposition 2.55. If X is spectrum with homotopy groups bounded below, then $\nu(X)$ is a filtered spectrum $F_\bullet$, and its limit is 0 and colimit is X. There is then a spectral sequence

$$E_2 \colon \pi_* C_* \Rightarrow \pi_* X,$$

where C_i is defined as the cofiber of $F_{i-1} \to F_i$.

Remark 2.56. There are also higher coherences present in this spectral sequence arising because $\nu(E)$ has some higher coherences in its construction. For example, the fact that $\mathrm{fil}^* \mathbb{S}$ is a ring produces the Leibniz rule.

2.3.2 The Adams–Novikov Spectral Sequence

Later in life, we will want to localize at a prime p.

Lemma 2.57. For each prime p,

$$\lim \tau_{\geq 2*}\,\mathrm{MU}_{(p)}^{\otimes \bullet} = \lim \tau_{\geq 2*}\,\mathrm{BP}_{(p)}^{\otimes \bullet + 1}.$$

Proof. Certainly there is a natural map $\mathrm{BP} \to \mathrm{MU}$, so we just have to check that it induces an isomorphism of the filtered spectra. By some argument with the Five lemma, it is enough to check that it gives an isomorphism on the associated graded spectra. Well, taking this grading passes through the limit, so it becomes a natural map

$$\Sigma^{2i} \lim \pi_{2i}\,\mathrm{BP}_{(p)}^{\otimes \bullet + 1} \to \Sigma^{2i} \lim \pi_{2i}\,\mathrm{MU}_{(p)}^{\otimes \bullet + 1},$$

which is now just a morphism taking place in $\mathrm{Mod}(\mathbb{Z})$. We can now use the Dold–Kan correspondence to pass to chain complexes. It turns out that the homology on the left computes cohomology of $\mathcal{M}_{\mathrm{fg},(p)}^{\mathrm{strict}}$ (which is the moduli stack of formal group laws up to strict isomorphism, and then we invert everything away from p), and the homology on the right computes moduli of p-typical formal group laws up to strict isomorphism. ■

Remark 2.58. In fact, the homology of the aforementioned complex is exactly the starting page of the spectral sequence.

Example 2.59. At $p = 2$, our MU complex looks like

$$\mathbb{Z}_{(2)}[v_1, \ldots] \to \mathbb{Z}_{(2)}[v_1, \ldots][t_1, \ldots] \to \mathbb{Z}_{(2)}[v_1, \ldots][t_1 \otimes t_1, \ldots].$$

For example, one can compute that $d(v_1) = \pm 2t_1$, which because $\mathbb{Z}_{(2)}$ is $\mathbb{Z}$-torsion-free, we see $d(t_1) = 0$. (Formulae for the differentials have been computed for many degrees.) One then sees that t_1 represents a class in the first homology which is 2-torsion. Unravelling the spectral sequence, one sees that an ith homology class in degree $2j$ (of the polynomial ring) maps to $\pi_{2j-i}\mathbb{S}$. For example, t_1 goes to the Hopf map in $\pi_1\mathbb{S}$.

Remark 2.60. Even though BP is not an $\mathbb{E}_\infty$-ring, it turns out that $\mathrm{fil}^* \mathbb{S}_{(p)} = \lim \tau_{\geq 2*}\mathrm{BP}^{\otimes \bullet + 1}$ is a filtered $\mathbb{E}_\infty$-ring, as we just showed. There are some reasons why $\mathrm{fil}^* \mathbb{S}_{(p)}$ may be "more canonical" than $\mathbb{S}$.

- One can find $\mathrm{fil}^* \mathbb{S}$ is the limit of the filtered spectra $\tau_{\geq 2*}E$ as E varies over all even $\mathbb{E}_\infty$-rings.
- The p-completed category $(\mathrm{SynSp})^\wedge_p$ embeds fully faithfully into the p-completed category of $\mathbb{C}$-motivic spectra.

One will frequently write down the E_2 page of our associated spectral sequence for $\lim \tau_{\geq 2*}\mathrm{BP}^{\otimes \bullet + 1}$ by graphing along axes i and $2j - i$, where i is the homological degree, and $2j$ is the top degree of the relevant polynomial in the homotopy groups of MU. (Recall that a class in homological degree i and polynomial degree $2j$ maps to $\pi_{2j-i}\mathbb{S}$.) For example, the E_2 page for $p = 2$ looks something like the following.

$$\begin{array}{cccc}
\vdots & \vdots & \vdots & \\
0 & 0 & t_1 \otimes t_1 \otimes t_1 & \cdots \\
0 & t_1 \otimes t_1 & ? & \cdots \\
t_1 & ? & ? & \cdots
\end{array}$$

Remark 2.61. It turns out that multiplication by t_1 is an isomorphism on the E_2 page, above a line of slope $1/5$. But $t_1^4 = 0$ (even though E_2 does not show this), it follows that the E_n page of the Adams–Novikov spectral sequence for large n (which can see $t_1^4 = 0$ vanishes) above some line of slope $1/5$.

Our next target is the following.

Theorem 2.62. For any positive slope, there is a page E_n which vanishes above a line of that slope.

One key idea in the proof will be to not merely view the spectral sequence as some formal algebraic gadget but as an actual object in the category of filtered spectra.

2.3.3 Towards Hopf Algebroids

We now see that we are interested in understanding the Čech nerve

$$\pi_*\mathrm{BP} \to \pi_*\mathrm{BP}^{\otimes 2} \to \pi_*\mathrm{BP}^{\otimes 3} \to \cdots.$$

This has the data of a "Hopf algebroid." Indeed, maps to R will get some kind of Agroup structure.

- A map $\pi_*\mathrm{BP} \to R$ is the data of a p-typical formal group law.

- The data of a map from
$$\pi_*\mathrm{BP}^{\otimes 2} = \pi_*\mathrm{BP} \otimes_{\pi_*\mathrm{BP}} \pi_*\mathrm{BP}$$
(note $\mathrm{BP} \otimes \mathrm{BP}$ is free as a module over $\pi_*\mathrm{BP}$) classifies two formal group laws with an isomorphism between them. For example, the canonical map from $\pi_*\mathrm{BP}$ explains what the identity is.
- The data of a map from $\pi_*\mathrm{BP}^{\otimes 3}$ similarly explains how to compose the two formal group laws from $\pi_*\mathrm{BP}^{\otimes 2}$.

In general, given any $\mathbb{E}_\infty$-ring R, if $R \otimes R$ is free over R, then one gets a Hopf algebroid.

2.4 February 25

Here we go.

2.4.1 Hopf Algebroids

Given an $\mathbb{E}_\infty$-ring R, one has an ∞-category of R-modules. If R is a discrete ring, then this is the derived category of R-modules. One may want such an explicit description of the R-modules in more general situations.

Definition 2.63 (totalization)**.** The *totalization* of a cosimplicial discrete ring $\Delta^{\mathrm{op}} \to \mathrm{Ring}$ is the limit of this functor.

Example 2.64. One can define a ring R as the limit of the Čech nerve
$$\mathrm{BP}_* \to \mathrm{BP}_*\mathrm{BP} \to \cdots .$$

Remark 2.65. Before totalization, a cosimplicial discrete ring is some "affine stack," meaning that it is a functor which takes a commutative ring to simplicial sets.

It will sometimes be true that such an affine stack will actually output to classical groupoids instead of arbitrary simplicial sets. In this situation, the cosimplicial ring is a Hopf algebroid.

Definition 2.66 (Hopf algebroid)**.** A *Hopf algebroid* is a pair of commutative rings (A, Γ) equipped with maps $\eta_L, \eta_R \colon A \to \Gamma$, a comultiplication $\Delta \colon \Gamma \to \Gamma \otimes_A \Gamma$, a counit $\varepsilon \colon \Gamma \to A$, and an inversion $\iota \colon \Gamma \to \Gamma$, all of which satisfy the following.

- η_L makes Γ into a flat A-module.
- There are various other identities so that $\mathrm{Hom}(A, R)$ is the set of objects and $\mathrm{Hom}(\Gamma, R)$ is the set of morphisms of a groupoid.

When no confusion is possible, we will simply call Γ the Hopf algebroid.

Remark 2.67. A Hopf algebroid (A, Γ) produces a cosimplicial ring as follows.
$$A \underset{\eta_R}{\overset{\eta_L}{\rightleftarrows}} \Gamma \;\substack{\longrightarrow\\ \longleftarrow\\ \longrightarrow\\ \longleftarrow\\ \longrightarrow}\; \Gamma \times_A \Gamma \qquad \cdots$$
The totalization is an $\mathbb{E}_\infty$-$\mathbb{Z}$-algebra.

Example 2.68. Any Hopf algebra Γ over a field k produces a Hopf algebroid by taking $\eta_L = \eta_R$ to both be the same identity map. For example, one can consider the Hopf algebra $\alpha_p := \mathbb{F}_p[x]/(x^p)$ over $\mathbb{F}_p$, which is a subscheme of $\mathbb{G}_{a,\mathbb{F}_p}$.

Example 2.69. If we take $A = \pi_*\mathrm{BP}$ and $\Gamma = \pi_*(\mathrm{BP}_*\mathrm{BP})$, then the Hopf algebroid (co)represents the moduli space of p-typical formal groups up to strict isomorphism.

We thus will satisfy ourselves with trying to understand the category of modules for such a Hopf algebroid. The point is to have some module for A and Γ which communicate with each other in some way.

Definition 2.70 (module)**.** Fix a Hopf algebroid Γ. Then a *discrete left Γ-comodule* is a discrete left A-module M equipped with an A-module map

$$\psi\colon M \to \Gamma \otimes_A M,$$

which is counital and coassociative.

Remark 2.71. The idea is that one can always present a module for the totalization as a sequence of modules, each for the discrete ring A and then Γ and then $\Gamma \otimes_A \Gamma$, and so on, always with some homotopies for coherence. But one can think of any module (indeed, any spectrum) by filtering it with its Postnikov tower, which in the case of $\mathbb{Z}$-modules allows one to recover chain complexes. The same thing works for Γ, thereby granting us our description of $\mathrm{Mod}(R)$, which are now the inductive systems of the bounded chain complexes. (Technically speaking, bounded chain complexes are those R-modules which are finite limits of just R, and we need to close this category under filtered colimits to get all of $\mathrm{Mod}(R)$.)

Remark 2.72. The (discrete) A-linear dual $\Gamma^\vee$ has the natural structure of a Hopf algebroid. Then the A-linear dual of a comodule M is simply a right $\Gamma^\vee$-module.

Remark 2.73 (Hahn)**.** Let's not pretend we care about homological algebra. Homological algebra is some fiddly way of pointing out modules.

2.4.2 Vanishing Lines

Let's return to α_p.

Example 2.74. Take $(A, \Gamma) = (\mathbb{F}_p, \mathbb{F}_p[x]/(x^p))$, and let R be the totalization. Then a perfect R-module is a bounded chain complex of finite-dimensional $\mathbb{F}_p$-modules M equipped with the structure map $\psi\colon M \to \mathbb{F}_p[x]/(x^p) \otimes_{\mathbb{F}_p} M$. However, such a map is the same as giving M the structure of an $\mathbb{F}_p[x]/(x^p)$-module because Γ is its own self-dual (indeed, $\alpha_p = \alpha_p^\vee$).

Example 2.75. We continue from the previous example. The homotopy groups of the totalization R of this Hopf algebroid can be computed to be $\mathrm{Ext}_\Gamma(\mathbb{F}_p, \mathbb{F}_p)$ via the Dold–Kan correspondence. This turns out to be $\mathbb{F}_p[x] \otimes \Lambda(h)$, where h is in degree 2 and β is in degree approximately p. Letting x have some grading (e.g., a $\mathbb{Z}/p\mathbb{Z}$-grading), then π_*R admits a bigrading, so we see that π_*R is supported on two lines of positive slope in the bigrading.

Warning 2.76. By convention, in all of our bigradings, homological degree is the vertical degree.

One can use the above examples to show the following.

Proposition 2.77. Let R be the totalization of the Hopf algebroid $\mathbb{F}_p[x]$, and choose a perfect R-module M. Then either

(a) $\pi_* M$ has a horizontal vanishing line in its bigraded homotopy groups, or

(b) there is a map $\Theta\colon \Sigma^{x,y} M \to M$ of R-modules which is not nilpotent and of the same slope as β. In this case, $M/\Theta \cong \operatorname{cofiber} \Theta$.

Proof. Simply classify all R-modules, as above. Roughly speaking, by the Dold–Kan correspondence, the homotopy groups of a discrete R-module M are computed as $\operatorname{Ext}_{\mathbb{F}_p[x]/(x^p)}(M, \mathbb{F}_p$. Thus, everything is built from the following two simple discrete modules.

- For the module $\mathbb{F}_p$, we see that its homotopy groups have the second property.
- For the module $\mathbb{F}_p[x]/(x^p)$, its homotopy groups are finitely supported. ■

One can even pass to higher powers.

Proposition 2.78. Let R be the totalization of the Hopf algebroid $\mathbb{F}_p[x]/\left(x^{p^2}\right)$. For every perfect R-module M, the homotopy groups $\pi_* M$ admit a vanishing line with minimal slope and an endomorphism Θ with slope parallel to that vanishing line. The cofiber of Θ will have a smaller slope vanishing line.

Proof. Note that R is an extension of the totalization R_0 of $\mathbb{F}_p[x]/(x^p)$ by itself. The point is that this sort of statement is immune to taking extensions in some formal way. ■

Here is another example.

Example 2.79. Let $\mathcal{P}_* := \pi_{2*}(\mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p)$ be the "even" Steenrod algebra, which is a free polynomial ring over $\mathbb{F}_p$ with generators ξ_i in degree $2p^i - 2$. It turns out that one has a Hopf algebra coming from $(A, \Gamma) := (\pi_{2*}\mathbb{F}_p, \mathcal{P}_*)$. It is difficult to explicitly write down the coproduct $\mathcal{P}_* \to \mathcal{P}_* \otimes_{\mathbb{F}_p} \mathcal{P}_*$, which come from the Adem relations. Nonetheless, we can define this map by taking homotopy groups of the composite

$$\mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p = \mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{S} \otimes_{\mathbb{S}} \mathbb{F}_p \to \mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p = (\mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p) \otimes_{\mathbb{F}_p} (\mathbb{F}_p \otimes_{\mathbb{S}} \mathbb{F}_p).$$

In particular, because $\mathbb{F}_p$ is a field, taking homotopy groups of the target produces $\mathcal{P}_* \otimes_{\mathbb{F}_p} \mathcal{P}_*$.

Remark 2.80. It turns out that the Hopf algebra $\mathcal{P}_*$ (co)represents the additive group with its automorphisms.

Remark 2.81. It turns out that the Hopf algebra $\mathcal{P}_*$ is an iterated extension of Hopf algebras, and this process gives us the associated graded which is a tensor product of $\mathbb{F}_p\left[\xi_i^{p^j}\right] / \left(\xi_i^{p^{j+1}}\right)$s.

2.5 March 2

Here we go.

2.5.1 Modules over Filtered Spectra

Let's talk some more about filtered spectra.

Definition 2.82 (bigraded sphere)**.** Let $S^{p,q}$ be the bigraded sphere, which is a filtered spectrum of the form

$$\cdots \to 0 \to 0 \to S^p \to S^p \to \cdots,$$

where the first S^p appears in degree $2q-p$. (Recall that degrees decrease going down the sequence.) In particular, $q \in \frac{1}{2}\mathbb{Z}$ is permitted.

Remark 2.83. All these bigraded spheres are $\otimes$-invertible, where $S^{0,0}$ is the unit.

Remark 2.84. With this grading convention, we see that $\Sigma S^{p,q} = S^{p+1,q+1/2}$.

Example 2.85. There is a double shift $\tau\colon S^{0,-1} \to S^{0,0}$, which looks like the following.

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d, equal] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & S^0 \arrow[r] \arrow[d, equal] & S^0 \arrow[r] \arrow[d, equal] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & S^0 \arrow[r] & S^0 \arrow[r] & S^0 \arrow[r] & S^0 \arrow[r] & \cdots
\end{tikzcd}$$

Notably, the first S^0 in the top occurs in degree -2.

Remark 2.86. If $X_\bullet$ is a filtered spectrum, then the bigraded homotopy groups $\pi_{*,*}X$ is a $\mathbb{Z}[\tau]$-module, and with our current grading conventions, one finds that τ shifts everything down by 2.

Recall that we are interested in the filtered $\mathbb{E}_\infty$-ring

$$\operatorname{fil}^* \mathbb{S}_{(p)} = \lim_{\Delta} \tau_{2\geq *} \operatorname{BP}^{\otimes(\bullet+1)}.$$

Taking the cofiber by τ passes to the associated graded, so one finds that

$$\pi_{2t-s,s}(\operatorname{fil}^* \mathbb{S}/\tau) = \operatorname{H}^s\left(\mathcal{M}^{\text{strict}}_{\text{fg},(p)}\right),$$

which is also $\operatorname{H}^s(\mathcal{M}_{\text{fg},(p)}; \omega^{\otimes t})$. (Recall that the formal groups up to strict isomorphism have an action by $\mathbb{G}_m$, which is where the extra grading comes from on the right-hand side.) On the other hand, inverting τ amounts to taking the colimit of the filtration, so we see that

$$\pi_{**} \operatorname{fil}^* \mathbb{S}_{(p)}\left[\tau^{-1}\right] = \pi_* \mathbb{S}_{(p)}.$$

It now turns out that τ^r-torsion elements correspond to d_{2r+1}-differentials: in each graded piece, there will be one element in the kernel and one element in the cokernel, which are the source and target for our differential.[1] In particular, if an element is going to live all the way to $\pi_*\mathbb{S}_{(p)}$, then it is not going to be τ-torsion and instead goes all the way down the complex.

Remark 2.87. Note that $\operatorname{fil}^* \mathbb{S}/\tau$ is an $\mathbb{E}_\infty$-ring: moving around the quotient, it is $\operatorname{fil}^* \mathbb{S} \otimes \left[S^{0,0}/\tau\right]$, and the latter is an $\mathbb{E}_\infty$-ring as the tensor product of $\mathbb{E}_\infty$-rings. We can thus consider modules over $\operatorname{fil}^* \mathbb{S}/\tau$, which turns out to be the derived category over $\mathcal{M}_{\text{fg},(p)}$. (The idea is that modding out by τ passes to the associated graded, which remains true "on the level of categories.") Notably, the elements of this category are filtered colimits of bounded chain complexes of $\operatorname{BP}_* \operatorname{BP}$ comodules. In particular, we have a large source of objects given by $\operatorname{BP}_* \operatorname{BP}$-comodules.

[1] The sort of example to have in mind is that $\tau\colon \mathbb{Z}[\tau]/\tau^r \to \mathbb{Z}[\tau]/\tau^r$ has one element in its kernel and cokernel.

Example 2.88. One has a comodule BP_* over the Hopf algebroid $\mathrm{BP}_*\,\mathrm{BP}$. Similarly, we have BP_*/p.

Non-Example 2.89. Recall that we can present BP_* as $\mathbb{Z}_{(p)}[v_1, v_2, \ldots]$. Then BP_*/v_1 fails to have a comodule structure because the right map $\mathrm{BP}_* \to \mathrm{BP}_*\mathrm{BP}$ is given by $v_1 \mapsto v_1 + pt_1$, so we cannot "just" take a quotient by v_1. However, the quotient $\mathrm{BP}_*/(p, v_1)$ is okay.

Example 2.90. In general, one has that $\eta_R(v_i) \equiv v_i \pmod{p, v_1, \ldots, v_{i-1}}$, so we are able to form the quotients $\mathrm{BP}_*/(p, v_1, v_2, \ldots)$. Notably, this is just $\mathbb{F}_p$.

Example 2.91. If X is any spectrum, then we have a comodule $\mathrm{BP}_{2*}X$. On homotopy, this looks like

$$\pi_{2*}(\mathrm{BP} \otimes X) \to \pi_{2*}(\mathrm{BP} \otimes \mathrm{BP} \otimes X) \to \cdots,$$

so the action by $\mathrm{BP}_*\,\mathrm{BP}$ is a little clearer.

In general, if we are handed a module over $\mathrm{fil}^*\,\mathbb{S}/\tau$, we can ask it comes from a module M over $\mathrm{fil}^*\,\mathbb{S}$. In particular, $M\left[\tau^{-1}\right]$ is then a spectrum which "lifts" its quotient M/τ.

Remark 2.92. Here is an open question: fix n, and take p large. Then we do not know if the module

$$\frac{\mathrm{BP}_*}{(p, v_1, \ldots, v_n)}$$

lifts to a spectrum. For example, even at $n = 1$, this is not possible if $p = 2$. This was recently proved for $n = 4$.

To continue, we need more module theory.

Definition 2.93 (thick subcategory)**.** Fix an $\mathbb{E}_\infty$-ring R. Given a collection $\{M_i\}_{i \in I}$ of R-modules, the *thick subcategory* of R-modules generated by these modules is the smallest full subcategory containing the modules and closed under finite limits, finite colimits, and retracts.

Definition 2.94 (perfect)**.** Fix an $\mathbb{E}_\infty$-ring R. Then a *perfect* module is an R-module in the thick subcategory generated by R.

2.5.2 Krause's Theorem

By extending the methods from last class, we can show the following.

Theorem 2.95 (Krause)**.** Fix a perfect module X over $\mathrm{fil}^*\,\mathbb{S}_{(p)}$. Then there is a slope

$$s \in \left\{ \frac{1}{p^{j+1}\left(p^i - 1\right)} : i > j \geq 0 \right\}$$

satisfying the following properties.

(a) $\pi_{*,*}X$ s concentrated below a line of slope s, and this slope s is minimal with this property.

(b) There is an endomorphism Θ of X with that slope such that $\operatorname{cofiber}\theta$ admits a smaller slope.

Example 2.96. Recall that there is a functor ν from spectra to filtered spectra which sends X to

$$\nu(X) := \lim \tau_{2*}\left(X \otimes \mathrm{BP}^{\otimes(\bullet+1)}\right).$$

If X is perfect, then $\nu(X)$ is perfect.

Remark 2.97. For any perfect X over $\mathrm{fil}^* \mathbb{S}_{(p)}$, the slope of X agrees with the slope of X/τ. Indeed, τ simply moves all the homotopy groups down, so slopes from X descend to X/τ. It also turns out that slopes for X/τ come from algebra (which follows from the proof) and thus lift to X.

Idea. By modding out by τ, we can reduce this to an algebra statement. In particular, for a perfect module M over $\mathrm{fil}^* \mathbb{S}/\tau$, we can form the quotient $M/(p, v_1, v_2, \ldots)$, and it is enough to prove the theorem for the slopes of this quotient.

However, the v_is basically shift the elements in $\pi_{**}M$ to the right. Let's now explain how this relates to what we discussed last class. Note that we can form the quotient $\mathrm{BP}_*\mathrm{BP}/(p, v_1, v_2, \ldots) = \mathbb{F}_p[t_1, t_2, \ldots]$. On the other hand, note that

$$\pi_*\left(\frac{\mathrm{BP} \otimes \mathrm{BP}}{(p, v_1, v_2, \ldots)}\right) = \pi_*(\mathbb{F}_p \otimes \mathrm{BP}) \to \pi_*(\mathbb{F}_p \otimes \mathbb{F}_p)^m,$$

where the second map is simply given by projecting BP_* to BP_0 via the Whitehead tower, which is $\mathbb{F}_p$. Now, this right-hand algebra is $\mathbb{F}[t_1, t_2, \ldots] \otimes \wedge^\bullet(\sigma t_1, \sigma t_2, \ldots)$, and it is the sort of thing we have some control over from last class: it is an iterated extension of some $\mathbb{F}_p[x]/\left(x^{p^r}\right)$s. ■

Remark 2.98. One can view Theorem 2.95 as a "purely algebraic" result. Indeed, it boils down to understanding the diagram

$$\begin{tikzcd} \mathrm{Mod}(\mathrm{fil}^* \mathbb{S}_{(p)}) \arrow[r, "/\tau"] \arrow[d, "\tau^{-1}"'] & \mathcal{D}(\mathcal{M}_{\mathrm{fg},(p)}) \\ \mathrm{Spectra}_{(p)} & \end{tikzcd}$$

as well as the full understanding of the category $\mathcal{D}(\mathcal{M}_{\mathrm{fg},(p)})$ (along with their slopes) produced last class.

2.5.3 The Main Theorem

We are now ready to prove Theorem 1.20.

Theorem 1.20 (Devinatz–Hopkins–Smith)**.** The E_∞ page of the Adams–Novikov spectral sequence lies under a curve which grows more slowly than any line.

Proof. By Theorem 2.95, there is a sequence of $\mathrm{fil}^* \mathbb{S}$-modules given by successively modding out by the produced endomorphisms $\Theta_1, \Theta_2, \ldots$. By inspection, any decreasing sequence in our set of slopes must go to zero. We can refine this choice of endomorphisms using Nishida nilpotence.

Example 2.99. Consider $\pi_{**} \mathrm{fil}^* \mathbb{S}_{(2)}$. Then our homotopy groups feature η in degree 1 which is not τ-torsion, and η would suffice to provide the endomorphism in slope 1. Additionally, η^2 is not τ-torsion, nor is η^3, but η^4 does vanish under τ. Thus, it is possible to choose this endomorphism Θ_1 to be τ-torsion.

Lemma 2.100 (Birkland)**.** Let the decreasing slopes of $\mathrm{fil}^* \mathbb{S}$ from Theorem 2.95 be $\{s_1, s_2, \ldots\}$. Then one can choose associated endomorphisms $\Theta_1, \Theta_2, \ldots$ of the correct slopes so that each Θ_i is τ-power torsion.

Proof. Example 2.99 explains how to choose Θ_1. Let's explain how we could choose Θ_2, and the remaining Θs are basically chosen by the same process. Then we are on the hunt from some endomorphism of $\mathrm{fil}^* \mathbb{S}/\Theta_1$ (up to the usual shifts giving the slope s_2). After inverting τ, we see that Θ_1 vanishes because it is τ-torsion, so taking this cofiber produces $\mathbb{S} \oplus \mathbb{S}$ (again, up to some suspensions).

Now, suppose that we use Theorem 2.95 to produce some endomorphism Θ_2. Then the key trick is that Theorem 1.10 tells us that $\tau^{-1}\Theta_2$ is a 2×2 matrix whose entries are nilpotent. It follows that $\tau^{-1}\Theta_2$ is nilpotent, so $\tau^a \Theta_2^b = 0$ for some a and b. Thus, we simply replace Θ_2 with Θ_2^b, and we are done! ■

Lemma 2.101. Choose τ-power torsion endomorphisms $\Theta_1, \Theta_2, \ldots$ of $\mathrm{fil}^* \mathbb{S}$. For each nonnegative integer n, the module $\mathrm{fil}^* \mathbb{S}$ is in the thick subcategory of $\mathrm{fil}^* \mathbb{S}$-modules generated by

$$\left\{ \frac{\mathrm{fil}^* \mathbb{S}}{\tau}, \frac{\mathrm{fil}^* \mathbb{S}}{\Theta_1, \ldots, \Theta_n} \right\}.$$

Proof. We proceed by induction; for $n = 0$, there is nothing to do. For $n \geq 1$, consider the cofiber sequence

$$\frac{\mathrm{fil}^* \mathbb{S}}{\Theta_1, \ldots, \Theta_n} \to \frac{\mathrm{fil}^* \mathbb{S}}{\Theta_1, \ldots, \tau\Theta_n} \to \frac{\mathrm{fil}^* \mathbb{S}}{\Theta_1, \ldots, \Theta_{n-1}, \tau}.$$

It follows that we can build the middle term from the two outer terms, so the middle term lives in the thick subcategory. Iterating this process allows us to build the module

$$\frac{\mathrm{fil}^* \mathbb{S}}{\Theta_1, \ldots, \tau^a \Theta_n}$$

for any a, so sending a large completes the proof by our induction. ■

Now, anything in the thick subcategory generated by these two modules admits a line of slope coming from the slope of $\mathrm{fil}^* \mathbb{S}/(\Theta_1, \ldots, \Theta_n)$, where everything above the line is τ-power torsion. It follows that the actual homotopy groups (on E_∞) must always live below this line! Letting n go to infinity, we see that the slopes go to zero. This completes the proof of Theorem 1.20. ■

2.6 March 4

Here we go.

2.6.1 The Master Theorem

Let's try to upgrade our proof of Theorem 1.20. Recall $\mathrm{SynSp} := \mathrm{Mod}(\mathrm{fil}^* \mathbb{S}_{(p)})$, and there is a functor $\nu \colon \mathrm{Spectra} \to \mathrm{SynSp}$ given by

$$\nu(X) := \lim \tau_{\geq 2*}(X \otimes \mathrm{BP}^{\otimes(\bullet+1)}).$$

Remark 2.102. If X is bounded below, then $\nu(X)$ is τ-complete, and $\nu X\left[\tau^{-1}\right] = X$. The moral is that we have been given a "motivic lift" from X as a spectrum to a synthetic spectrum.

Remark 2.103. It turns out that ν commutes with filtered colimits of bounded below spectra. However, it does not preserve cofibers: the problem is that a map $f\colon X \to Y$ produces a sequence

$$\tau_{2*}X \to \tau_{2*}Y \to \tau_{2*}\operatorname{cofiber}(X \to Y)$$

which is not expected to be a cofiber sequence. In particular, one has an issue with the numerology of the cutoff, so the difference from the above being a cofiber sequence is just given by an Eilenberg–MacLane space. As a result, one finds that the cofiber of the sequence

$$\operatorname{cofiber}(\nu f) \to \nu \operatorname{cofiber}(f)$$

is τ-torsion.

Theorem 2.104. There is a curve C in the plane with the following property: for any X with $\pi_i X = 0$ for all $i < 0$, then $\pi_{**}(\nu X)$ is τ-power-torsion above the curve. Additionally, all lines of constant slope are eventually above C.

Proof. Any such X is a filtered colimit of spheres, so this follows from properties of ν and Theorem 1.20. In particular, even though ν does not commute with cofibers, it does commute with them up to some τ-torsion, which we notably do not care about for the application. ■

Professor Hahn called Theorem 2.104 the "master theorem," from which we will derive many other results of Devinatz–Hopkins–Smith.

Remark 2.105. Let I be the fiber of $\mathbb{S}_{(p)} \to \mathrm{BP}$. If X is bounded below, then there is a spectral sequence associated to

$$\cdots \to I^{\otimes 2} \otimes X \to I \otimes X \to X \to X \to X \to \cdots.$$

(The spectral sequence converges because I is concentrated in positive degrees, so higher tensor-powers of I only live in high degrees.) It turns out that the E_2-page of this spectral sequence recovers the one from $\pi_{**}\operatorname{fil}^* \mathbb{S}/\tau$. The moral is that this gives an interpretation of the x-axis of our Adams–Novikov spectral sequence: on the E_∞-page, the x-axis of $\pi_{**}\nu X$ is the image of $\pi_* X \to \mathrm{BP}_* X$!

This interpretation of the Adams–Novikov spectral sequence has the following application.

Corollary 2.106. Let R be a p-local connective $\mathbb{E}_1$-ring (meaning that R has no negative homotopy groups). Then every class in

$$\ker(\pi_* R \to \mathrm{BP}_* R)$$

is nilpotent.

Proof. Let's look at the E_∞-page of $\nu(R)$. Then by Remark 2.105, every class in the kernel of the given map is a class strictly above the x-axis. But all such classes are nilpotent by Theorem 2.104. ■

Remark 2.107. This is a first indication that Theorem 1.20 is going to be powerful: we have managed to prove something about rather general rings.

We can actually upgrade Corollary 2.106 to work with arbitrary $\mathbb{E}_1$-rings. This requires a trick.

Definition 2.108 (smash nilpotent)**.** Fix a p-local spectrum X. A class $\alpha \in \pi_i X$ is *smash nilpotent* if and only if $\alpha^{\otimes N}\colon S^{iN} \to X^{\otimes N}$ is nullhomotopic for some N.

Proposition 2.109. Fix a spectrum X and a class $\alpha \in \pi_i X$. Then α is smash nilpotent if $\mathrm{MU} \otimes \alpha$ is nullhomotopic.

Proof. Note that any spectrum is filtered colimit of perfect spectra, so we can present X as a filtered colimit of $\mathbb{S}_{(p)}$-modules, which we denote $X = \operatorname{colim}_i X_i$. Because S^i is compact, we see that S^i factors through one of those X_i. We may even factor the nullhomotopy of $\mathrm{MU} \otimes \alpha$ through one of the X_is. Thus, we may reduce to the case where X is a perfect $\mathbb{S}_{(p)}$-module. Because X is now finitely constructed from $\mathbb{S}_{(p)}$s, it is bounded below, so $\Sigma^d X$ is connective for sufficiently large d. Furthermore, $\Sigma^d \alpha \in \pi_{i+d} \Sigma^d X$ will be nilpotent if and only if α is.

We are thus basically reduced to the case where X is connective. Note that we can form an $\mathbb{E}_1$-ring

$$\mathbb{S} \oplus \Sigma^d X \oplus \left(\Sigma^d X\right)^{\otimes 2} \oplus \cdots .$$

(This is the free $\mathbb{E}_1$-ring on $\Sigma^d X$.) Being smash nilpotent is simply the claim that the image of $\Sigma^d \alpha$ in this $\mathbb{E}_1$-ring is nilpotent. But our class is now nilpotent in this ring by Corollary 2.106. ■

Corollary 2.110. Let R be a p-local $\mathbb{E}_1$-ring. Then every class in

$$\ker(\pi_* R \to \mathrm{BP}_* R)$$

is nilpotent.

Proof. Any element in the kernel of $S^i \to R$ is smash nilpotent. Taking a power is then nullhomotopic, so the original element must have been nilpotent. ■

Example 2.111. Let R be a p-local $\mathbb{E}_1$-ring. Then R vanishes if and only if 1 is nilpotent, which is then true if and only if 1 is in the kernel of the map $\pi_* R \to \mathrm{BP}_* R$. Unwinding, we see that R vanishes if and only if $\mathrm{BP}_* R$ vanishes.

2.6.2 The Spectrum MUP

For our next application, we want the following construction.

Definition 2.112. Let MUP be the Thom spectrum of the canonical map $\mathrm{BU} \times \mathbb{Z} \to \operatorname{Pic} \mathbb{S}$.

Remark 2.113. It turns out that $\pi_* \mathrm{MUP} = \mathbb{Z}[\{x_i\}][u, 1/u]$, where u lives in degree 2, and x_i lives in degree $2i$.

Recall that we can construct $\operatorname{Pic} \mathrm{MUP}$, which is the subcategory of $\otimes$-invertible modules, equipped with only automorphisms as morphisms. There is a full subcategory $\mathrm{BGL}_1 \mathrm{MUP} \subseteq \operatorname{Pic} \mathrm{MUP}$ above MUP. For example,

$$\pi_i \mathrm{BGL}_1 \mathrm{MUP} \subseteq \operatorname{Pic} \mathrm{MUP} = \begin{cases} 0 & \text{if } i = 0, \\ \pi_0 \mathrm{MUP}^\times & \text{if } i = 1, \\ \pi_{i-1} \mathrm{MUP} & \text{if } i \geq 2. \end{cases}$$

The case of $i = 0$ has no content, the case of $i \geq 2$ follows because the higher homotopies are all invertible in any ∞-category anyway. The case of $i = 1$ holds because we are hunting for invertible morphisms.

Example 2.114. A map $f\colon S^3 \to \mathrm{BGL}_1\,\mathrm{MUP}$ is given by an element in $\pi_2\mathrm{MUP}$; for concreteness, one could take $u \in \pi_*\mathrm{MUP}$. It turns out that $\mathrm{Thom}(f)$ is some colimit of S^3, and by imagining S^3 as gluing together the two caps, we find that the Thom spectrum is

$$\mathrm{Thom}(f) = \mathrm{cofiber}\left(f\colon \Sigma^2\mathrm{MUP} \to \mathrm{MUP}\right).$$

Here is something amusing: S^3 can be realized as $\mathrm{SU}(2)$, so one finds that $S^3 = \Omega\mathbb{HP}^\infty$. Then f lifts to a map $S^4 = \Sigma S^3 \to \mathrm{B}^2\,\mathrm{GL}_1\,\mathrm{MUP}$. One may hope to extend this map from S^4 all the way to $\mathrm{HP}^\infty \to \mathrm{B}^2\,\mathrm{GL}_1(\mathrm{MUP})$, which is possible because the only obstructions to gluing would live in the odd homotopy groups of the target, which are the even homotopy groups of MUP, which all vanish. Thus, f arises from loops of some map $\mathbb{HP}^\infty \to \mathrm{B}^2\,\mathrm{GL}_1(\mathrm{MUP})$, so the cofiber $\mathrm{Thom}(f) = \mathrm{MUP}/f$ is an $\mathbb{E}_1$-algebra! Because the lifting is not unique, the $\mathbb{E}_1$-structure is not unique.

Remark 2.115. In fact, one can iterate the previous example for any sequence of elements $f_1, f_2, \ldots \in \pi_2\mathrm{MUP}$ to show that $\mathrm{MUP}/(f_1, f_2, \ldots)$ is an $\mathbb{E}_1$-algebra. For example, the quotient

$$R := \frac{\mathrm{MUP}}{(2u, x_1 - u, x_2u^{-1}, x_3u^{-2}, \ldots)}$$

admits the structure of an $\mathbb{E}_1$-algebra and has homotopy groups $\pi_*R = \mathbb{F}_2[x_1, 1/x_1]$, so it is a skew field.

Thus, we have produced a rather interesting example of a skew field. We will shortly classify all of them.

2.7 March 9

Today we continue with our applications of slopes.

Remark 2.116. Fix an odd prime p. The element v_1 (constructed on the homework) corresponds to a map $v_1\colon \Sigma^{2p-2}\mathbb{S}/p \to \mathbb{S}/p$. For example, we see that $(\mathbb{S}/p)_*(X)$ is an $\mathbb{F}_p[v_1]$-module for any X.

2.7.1 Morava K-Theory

Let's start by explaining some more general quotients from last class.

Proposition 2.117. Suppose that R is an $\mathbb{E}_2$-ring for which π_*R is supported in even degrees. For any $x_1, \ldots, x_n \in R$, the quotient $R/(x_1, \ldots, x_n)$ can be given the structure of an $\mathbb{E}_1$-R-algebra.

Proof. For $R = \mathrm{MUP}$, we already saw how to do this for π_2 in Remark 2.115. The same argument works for any R for our elements in π_2.

In general, there is a product decomposition $\mathrm{BU} = \mathbb{CP}^\infty \times \mathrm{BSU}$. (Thinking of BU as virtual vector bundles, one has a leftward map by taking actual line bundles, and one has a map from the right to the left inverting it.) Applying Ω^2, we see that

$$\Omega^2\mathrm{BU} = \mathbb{Z} \times \mathrm{BU}$$

as 2-fold loop spaces (by Bott periodicity). Now, one can take the Thom spectrum of the functor $\mathbb{N} \xrightarrow{m} \mathbb{Z} \times \mathrm{BU} \to \mathrm{Pic}\,\mathbb{S}$ to produce $\mathbb{E}_2$-rings $\mathbb{S}[x_{2m}]$. One then needs to solve some lifting property as in Remark 2.115, which eventually follows from the fact that the even-ness of R and the fact that $\mathbb{CP}^\infty$ has a cell structure using only cells of even dimension. ∎

Remark 2.118. The lifting described at the end of the argument implies that the $\mathbb{E}_1$-structure is far from canonical.

Here is our application.

Definition 2.119. Recall that $\pi_*\mathrm{BP} = \mathbb{Z}_{(p)}[v_1, v_2, \ldots]$, where v_i lives in degree $2p^i - 2$. For each $n \geq 0$, we define $\mathrm{BP}\langle n\rangle$ as the quotient

$$\frac{\mathrm{BP}}{(v_{n+1}, v_{n+2}, \ldots)}.$$

Remark 2.120. By construction, we see that $\mathrm{BP}\langle n\rangle$ admits the structure of an $\mathbb{E}_1$-BP-algebra, though it does not even admit a unique structure as a BP-module because that requires a choice of generators $v_\bullet$s!

Remark 2.121. For $p > 2$ or when $p = 2$ and $n \leq 2$, it turns out that $\mathrm{BP}\langle n\rangle$ is unique as a spectrum. In these cases, it is reduced to some Ext calculation, which is still open at $p = 2$.

We can further pass to "residue fields."

Definition 2.122. We further define $k(n) \coloneqq \mathrm{BP}\langle n\rangle/(p, v_1, \ldots, v_{n-1})$ and $K(n) \coloneqq v_n^{-1}\mathrm{BP} \otimes_{\mathrm{BP}} k(n)$.

Remark 2.123. Note that $\pi_* k(n) \cong \mathbb{F}_p[v_n]$, where v_n lives in degree $2p^n - 2$. It is known that $k(n)$ is unique as a spectrum.

Remark 2.124. By construction, we see that $\pi_* K(n) = \mathbb{F}_p[v_n, 1/v_n]$, so $K(n)$ is a skew field. In particular, every $K(n)$-module is free.

Remark 2.125. The $K(n)$s are called Morava K-theory. It turns out that $K(1)$ appears in the decomposition

$$\mathrm{KU}/p = K(1) \oplus \Sigma^2 K(1) \oplus \cdots \oplus \Sigma^{p-1} K(1),$$

where we will choose not to define KU.

2.7.2 Characteristics of Skew Fields

We are now ready to classify characteristics of skew fields. To start, we note that we don't need to remove the lower vs to get a free module.

Proposition 2.126. The spectrum

$$\left(\frac{\mathrm{BP}}{(p, v_1, \ldots, v_{n-1})}\right)[1/v_n]$$

is an infinite direct sum of shifts of $K(n)$.

Proof. Recall that there is a map $\eta_R \colon \mathrm{BP} \to \mathrm{BP} \otimes \mathrm{BP}$. On the other hand, $\pi_*(\mathrm{BP} \otimes \mathrm{BP})$ was a polynomial ring over $\pi_*\mathrm{BP}$, which we saw from a decomposition

$$\mathrm{BP} \otimes \mathrm{BP} = \bigoplus_{t_I} \Sigma^{\deg t_I} \mathrm{BP}.$$

Thus, we have a composite

$$\mathrm{BP} \to \mathrm{BP} \otimes \mathrm{BP} = \bigoplus_{t_I} \Sigma^{\deg t_I} \mathrm{BP} \twoheadrightarrow \bigoplus_{\text{monomials } t_I \in \pi_* \mathrm{BP}[t_1^{p^n}, t_2^{p^n}, \ldots]} \Sigma^{\deg t_I} \mathrm{BP}\langle n \rangle,$$

where the last map is a projection. Modding out by $(p, v_1, \ldots, v_n)$ and then inverting v_n produces a map from $\left(\frac{\mathrm{BP}}{(p, v_1, \ldots, v_{n-1})}\right)[1/v_n]$ to a sum of shifts of $K(n)$. One can then compute this composite on homotopy groups, where it is seen to be an isomorphism. ■

Here is a useful notion for classification.

Definition 2.127 (Bousfield class)**.** Two spectra E and F are in the same *Bousfield class* if and only if $E \otimes X = 0$ is equivalent to $F \otimes X = 0$ for all spectra X.

Proposition 2.128. The spectrum BP is in the same Bousfield class as

$$\mathbb{F}_p \oplus \mathbb{Q} \oplus K(1) \oplus K(2) \oplus \cdots.$$

Proof. We will use the following claim repeatedly: for any spectrum E with endomorphism $\nu \colon \Sigma^k E \to E$, one has that E and $E/\nu \oplus E[1/\nu]$ are in the same Bousfield class. Indeed, to see why this completes the proof, note that $\mathrm{BP} \otimes X$ vanishes if and only if both $\mathrm{BP}[1/p] \otimes X$ and $\mathrm{BP}/p \otimes X$ vanishes. The former is equivalent to $\mathbb{Q} \otimes X$. Then the latter is equivalent to having $\mathrm{BP}/p[1/v_1] \otimes X = 0$ and $\mathrm{BP}/(p, v_1) \otimes X = 0$. The former is equivalent to $X \otimes K(1) = 0$, and for the latter we keep going to $K(2)$, then $K(3)$, and so on. The $\mathbb{F}_p$ appears because one should show that $X \otimes \mathbb{F}_p$ vanishes if and only if $X \otimes K(n)$ vanishes for n large enough. Indeed,

$$\mathbb{F}_p = \frac{\mathrm{BP}}{(p, v_1, v_2, \ldots)},$$

which is some filtered colimit, so having $X \otimes \mathbb{F}_p$ vanish means that it will vanish after some finite colimit, meaning that $X \otimes K(n)$ vanishes for n large enough.

It remains to show the claim. Certainly $E \otimes X$ vanishing implies that $E/\nu \otimes X$ and $E[1/\nu] \otimes X$ vanishes. Conversely, $E/\nu \otimes X$ vanishing means that the map $\nu \colon \Sigma^k E \otimes X \to E \otimes X$ is an equivalence, so ν acts by an isomorphism on $\pi_*(E \otimes X)$. But $E[1/\nu] \otimes X$ vanishes, so this isomorphism must vanish, so $\pi_*(E \otimes X) = 0$, so $E \otimes X = 0$. ■

Remark 2.129. Thus, by Example 2.111, we see that a p-local $\mathbb{E}_1$-ring R vanishes if and only if $R \otimes \mathbb{F}_p = 0$ or $R \otimes \mathbb{Q} = 0$ or $R \otimes K(n) = 0$ for all $n \geq 1$. Presumably, it is easier to check if tensoring with one of the fields $\{\mathbb{F}_p, \mathbb{Q}\} \cup \{K(n)\}_{n \geq 1}$ vanishes is easier because one can use Künneth formulae and the like.

We are now ready to classify some fields.

Definition 2.130 (characteristic)**.** Two skew fields R_1 and R_2 are in the same *characteristic* if and only if $R_1 \otimes R_2 \neq 0$.

One can check that this is an equivalence relation, though it requires a little effort.

Remark 2.131. It turns out that the skew fields $\{\mathbb{Q}, \mathbb{F}_p\} \cup \{K(n)\}_{n \geq 1}$ all live in different characteristic classes.

Proposition 2.132. Every p-local skew field has the characteristic of $\mathbb{Q}$, $\mathbb{F}_p$, or $K(n)$ for some $n \geq 1$.

Proof. Let R be a p-local skew field. Then R is nonzero, so Remark 2.129 tells us that some element in

$$\{R \otimes \mathbb{Q}, R \otimes \mathbb{F}_p\} \cup \{R \otimes K(n)\}_{n \geq 1}$$

is nonzero, so the statement follows. ■

Remark 2.133. One can remove the p-local hypothesis by simply letting p vary.

2.7.3 Finite Spectra

Let's next talk about finite spectra.

Definition 2.134 (finite)**.** A *finite spectrum* is a perfect $\mathbb{S}_{(p)}$-module.

Here are some quick facts.

Lemma 2.135. Fix a finite spectrum X. Then X vanishes if and only if $X \otimes \mathbb{F}_p = 0$.

Proof. For abelian groups, this is basically Nakayama's lemma. One then combines this with the Hurewisz theorem in order to show that $X \otimes \mathbb{F}_p = 0$ implies that we can kill the homotopy groups one at a time. ■

Lemma 2.136. Fix a nonzero finite spectrum X. Then $K(n)_*X \neq 0$ for n sufficiently large.

Proof. There is an Atiyah–Hirzebruch spectral sequence

$$\mathrm{H}_*(X; \mathbb{F}_p[v_n, 1/v_n]) \Rightarrow K(n)_*X.$$

Now, because X is finite, $\pi_*(X \otimes \mathbb{F}_p) = \mathrm{H}_*(X; \mathbb{F}_p)$ is concentrated in only finitely many degrees (and is nonzero by Lemma 2.135). The left-hand side is some sum of $\Sigma^{m(2p^n-2)}\mathrm{H}_*(X; \mathbb{F}_p)$s, so there is a finite band in the plane of the spectral sequence in which the E_2-page concentrates. But as n goes large, the differentials will be unable to be nonzero, so the spectral sequence immediately converges, and we see that $K(n)_*X$ is nonzero. ■

Lemma 2.137. Fix a finite spectrum X. If $K(n) \otimes X = 0$, then $K(n-1) \otimes X = 0$.

Proof. Consider the $\mathbb{E}_1$-ring

$$R = \left(\frac{\mathrm{BP}}{(p, v_1, \ldots, v_{n-2}, v_{n+1}, v_{n+2}, \ldots)} \right) [1/v_n]$$

so that $\pi_*R = \mathbb{F}_p[v_{n-1}, v_n, 1/v_n]$. Because X is perfect, $\pi_*(R \otimes X)$ is a finitely generated module over π_*R, and it vanishes $\pmod{v_{n-1}}$. But π_*R is some kind of polynomial ring in one variable over a graded field, so such (graded) modules can be classified. Using this classification, one is able to see that vanishing $\pmod{v_{n-1}}$ implies that $\pi_*(R \otimes X)[1/v_{n-1}]$ also vanishes. Indeed, R_*X is $(2p^n-2)$-periodic (because v_n has invertible action), and it is a module over π_0R, which is $\mathbb{F}_p[t]$ where $t = v_{-1}^d v_n^{-1}$ for some d. As such, we are looking at $2p^n - 2$ modules over $\mathbb{F}_p[t]$, and our conditions that we vanish $\pmod{v_{n-1}}$ and vanish after inverting v_{n-1} gather together to show that our module over a principal ideal domain implies that $K(n-1) \otimes X = 0$. ■

Combining these lemmas grants us an invariant.

Definition 2.138 (type)**.** Fix a finite spectrum X. Then the *type* of X is the unique nonnegative integer n for which $K(m) \otimes X = 0$ for $m < n$ and $K(m) \otimes X \neq 0$ for $m \geq n$. By convention, $K(0) = \mathbb{Q}$.

Example 2.139. One has that $\mathbb{S}_{(p)}$ is type 0 because $\mathbb{S}_{(p)} \otimes \mathbb{Q} \neq 0$.

Example 2.140. Note that $\mathbb{S}/p$ is type 1. It is not hard to see that $\mathbb{Q} \otimes \mathbb{S}/p = 0$, and one can check that $K(1) \otimes \mathbb{S}/p \neq 0$.

Example 2.141. For odd primes p, we showed on the homework that the finite spectrum $\mathbb{S}/(p, v_1)$ is type 2.

Example 2.142. Let η be the Hopf fibration. Then $\mathbb{S}/\eta$ is type zero: η is a map $\Sigma\mathbb{S} \to \mathbb{S}$, so $\mathbb{Q} \otimes \mathbb{S}/\eta$ is the cofiber of some map $\Sigma\mathbb{Q} \to \mathbb{Q}$. But all such maps vanish, so $\mathbb{Q} \otimes \mathbb{S}/\eta = \mathbb{Q} \oplus \Sigma^2\mathbb{Q}$, which is nonzero.

Remark 2.143. It was verified (by brute force construction) that there is a finite spectrum of each type. One basically takes finite retracts of tensor products of some well-understood "lens spaces."

2.8 March 11

Here we go.

2.8.1 Thick Subcategories of Finite Spectra

To convince ourselves that the type is a good invariant on our category, we should check that it has good categorical properties.

Theorem 2.144. For each $n \geq 0$, the category $\mathcal{C}_{\geq n}$ of finite spectra of type at least n is a thick subcategory of $\mathbb{S}_{(p)}$-modules. Conversely, any thick subcategory of finite spectra is $\{0\}$ or takes the form $\mathcal{C}_{\geq n}$ for some $n \geq 0$.

Example 2.145. It would follow from Example 2.142 and Theorem 2.144 that the thick subcategory generated by $\mathbb{S}/\eta$ should have all finite spectra. Well, $\mathbb{S}/\eta^2$ is an extension of $\mathbb{S}/\eta$ by itself, and one can similarly construct $\mathbb{S}/\eta^3$ and $\mathbb{S}/\eta^4$. But $\eta^4 = 0$, so $\mathbb{S}/\eta^4 = \mathbb{S} \oplus \Sigma^5\mathbb{S}$, which contains $\mathbb{S}$ as a retract. Thus, we get all perfect $\mathbb{S}$-modules! Note that this example suggests that we will have to use some machinery: it is not obvious that all elements in $\pi_*\mathbb{S}$ are nilpotent!

Proof of Theorem 2.144. It is not too hard to check that $\mathcal{C}_{\geq n}$ is actually thick. This is a matter of checking that the type can only increase when taking various finite (co)limits and retracts and so on.

Thus, the hard part is to show the following: if $\mathcal{C}$ is a thick subcategory containing a finite spectrum of type n, then $\mathcal{C}$ contains every finite spectrum of type at least n. Before proceeding with the argument, we establish some notation.

Definition 2.146 (dual)**.** For a finite spectrum X, we define the *dual* to be

$$DX := \mathrm{Hom}_{\mathrm{Spectra}}(X, \mathbb{S}_{(p)}).$$

Remark 2.147. Note that DX is a finite spectrum: certainly $D\mathbb{S}_{(p)} = \mathbb{S}_{(p)}$ is a finite spectrum, and general X can be built out of retracts of finite (co)limits, which commute with Hom.

Remark 2.148. Using similar arguments to the previous remark, one can check that the canonical map

$$X_1 \otimes DX_2 \to \mathrm{Hom}_{\mathrm{Spectra}}(X_2, X_1)$$

is an isomorphism.

Notation 2.149. For a finite spectrum X, we also define $\mathrm{End}\, X := X \otimes DX$.

Remark 2.150. We note that $\mathrm{End}\, X$ is an $\mathbb{E}_1$-ring.

We now proceed with the argument.

1. We now proceed with the proof. Fix our $X \in \mathcal{C}$ of type n. Then $\mathrm{End}\, X := X \otimes DX$ is an $\mathbb{E}_1$-ring, so it admits a unit $e\colon \mathbb{S}_{(p)} \to \mathrm{End}\, X$, whose fiber we denote by $f\colon F \to \mathbb{S}_{(p)}$. Thus, we have built a fiber sequence
$$F \xrightarrow{f} \mathbb{S}_{(p)} \xrightarrow{e} \mathrm{End}\, X.$$
Because $K(m)_* X \neq 0$ for $m \geq n$, we see that $K(m)_* e$ is injective: this amounts to asserting that the target ring $K(m)_* \mathrm{End}\, X$ is nonzero! The long exact sequence thus implies that $K(m)_* f = 0$ for $m \geq n$.

2. Now, choose some Y of type at least n, and we would like to check that Y is in $\mathcal{C}$. We still have a unit map $\mathbb{S}_{(p)} \to Y \otimes DY$, so the composite
$$F \to \mathbb{S}_{(p)} \to \mathrm{End}\, Y$$
vanishes on $K(m)_*$ for $m \geq n$ still by the previous paragraph. Note that this implies that the composite vanishes on $\mathbb{F}_{p*}$ as well: $\mathbb{F}_p$ is some infinite colimit of the $K(m)$s, so because Y is finite, the $\mathbb{F}_p$-homology will agree with $K(m)$-homology for m large enough.[2] We also remark that the composite vanishes on $K(m)_*$ for $m < n$ because $K(m)_* Y = 0$ for $m < n$ already.

3. We now use our machinery. By Corollary 2.110 and Proposition 2.128, we conclude that there is a k large enough for which the composite
$$F^{\otimes k} \to (\mathrm{End}\, Y)^{\otimes k} \to \mathrm{End}\, Y$$
vanishes. By moving the duals around, we can show that
$$F^{\otimes k} \otimes Y \xrightarrow{f} F^{\otimes (k-1)} \otimes Y \xrightarrow{f} \cdots \xrightarrow{f} Y$$
is trivial for some k.

4. We complete the proof. By the previous step, we conclude that Y is a retract of $Y/(F^{\otimes k} \otimes Y)$ by the long exact sequence in homotopy. Splitting up this large tensor product, we see that Y is in the thick subcategory generated by $(F^{\otimes a} \otimes Y)/(F^{\otimes (a+1)} \otimes Y)$s, but this latter object is $F^{\otimes a} \otimes Y \otimes DX$. This last object is a tensor product of X with a finite spectrum, which is in $\mathcal{C}$: indeed, tensor products commute with finite limits, colimits, and retracts, so this tensor product can be turned into a retract of some finite (co)limit of X. ■

[2] This is a little tricky: imagine taking $- \otimes K(m)$, and because $K(m)$ is a field, everything splits, and one can try hard to compute with a spectral sequence.

2.8.2 Self Maps of Finite Spectra

The following example motivates our next theorem.

Example 2.151. Note that $\mathbb{S}$ has type 0, and it has a self map $p\colon \mathbb{S} \to \mathbb{S}$ with cofiber of type 1. Similarly, $\mathbb{S}/p$ is type 1, and it has a self map with cofiber of type 2: for $p > 2$, we can take $v_1\colon \mathbb{S}^{2p-2}/p \to \mathbb{S}/p$; for $p = 2$, we can take $v_1^4\colon \mathbb{S}^8/2 \to \mathbb{S}/2$.

Let's codify a good property of these self-maps.

Definition 2.152. Fix a finite spectrum X. Then a v_n*-self map* is an endomorphism $f\colon \Sigma^K X \to X$ such that $K(n)_* f$ is an isomorphism, but $K(m)_* f$ is nilpotent for $m \neq n$.

These maps all exist.

Theorem 2.153. Let X be a finite spectrum of type n. Then X admits a v_n-self map $\Sigma^k X \to X$.

Remark 2.154. This is intended to be a little similar to our story with slopes: given a particular slope, we have a self-map, and taking the quotient by a self-map should give a smaller slope.

Remark 2.155. Note that the cofiber of a v_n-self map automatically has type $n+1$, which can be seen by taking $K(n)_*$ of the corresponding fiber sequence.

Before proceeding with the proof, we need a bit more language.

Definition 2.156. Fix a finite p-local ring $\mathbb{E}_1$-ring R. Then a v_n*-element* in $\pi_* R$ is a homotopy class which maps to a unit in $K(n)_* R$ and a nilpotent element in $K(m)_* R$ for $m \neq n$.

Remark 2.157. By unwinding the definitions, we see that a v_n-self map of X is a v_n-element in $\pi_*(X \otimes DX)$.

The following lemma explains the terminology.

Lemma 2.158. Let x be a v_n-element of a finite p-local ring R. Then there is N such that $K(m)_* x^N = 0$ for $m \neq n$ and such that $K(n)_* x$ is a power of v_n.

Proof. Everything is perfect, so the homology is given by just $\mathbb{F}_{p*}$ for m large enough, so we just choose N large enough so that $K(m)_* x^N = 0$ for m small and $\mathbb{F}_{p*} x^N = 0$. As for the second requirement, note that $K(n)_* R/(v_n - 1)$ is a finite set (we are kind of looking at the group of units of a finite ring), so this is just a congruence condition on N. ■

These v_n-elements turn out to have a lot of structure. To begin, we will also want the following piece of algebra.

Lemma 2.159. Let A be a discrete associative ring. Suppose that $x, y \in A$ are chosen such that $xy = yx$ and such that $x - y$ is p-power torsion and nilpotent. Then $x^{p^N} = y^{p^N}$ for N large enough.

Proof. Write x^{p^N} as

$$(y + (x - y))^{p^N},$$

and we imagine expanding this out using the binomial theorem. There is always a "main term" y^{p^N}, and the remaining terms look like $\binom{p^N}{d} y^{p^N-d}(x-y)^d$. For larger d, this automatically vanishes by nilpotence, and for large enough N, we can make the smaller terms have $\binom{p^N}{d}$ divisible by a high power of p. ■

Here is our application of the algebra.

Lemma 2.160. Let R be a finite p-local ring, and suppose that x is a v_n-element in $\pi_* R$ (for some positive n). Then some power of x is central in $\pi_* R$.

Proof. We consider the ring $R_2 := X \otimes DX$. Now, R_2 has two elements a and b which are given by left- and right-multiplication by x. By Lemma 2.159, it is enough to check that a and b commute, have difference which is torsion, and $a - b$ is nilpotent. The commutativity has little content (because left- and right-multiplication should always commute), and being torsion follows because R is finite (namely, everything has trivial homology over $\mathbb{Q}$). Lastly, $a - b$ is nilpotent by Lemma 2.158: it is enough to check the vanishing for every $K(n)$, and the lemma tells us that a high power of a and b acts the same on both sides, so the nilpotence basically follows. ■

Proposition 2.161. Let R be a finite p-local ring, and suppose that x and y are v_n-elements. Then $x^a = y^b$ for some a and b.

Proof. By Lemma 2.160, we may replace x and y with powers which commute. Then the same argument as in Lemma 2.160 shows that a high enough power of x and y must be equal. ■

Corollary 2.162. Let $f\colon X \to Y$ be a morphism of finite spectra. If X and Y admit v_n-self maps g and h, then there is a commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "g^a"'] & Y \arrow[d, "h^b"] \\ \Sigma^k X \arrow[r, "f"] & \Sigma^k Y \end{tikzcd}$$

for some a and b.

Proof. Both h and Dg are v_n-self maps of $Y \otimes DX$, so some power of them agree by Proposition 2.161. The given equality follows by evaluating 1 on the composite

$$\mathbb{S} \xrightarrow{f} Y \otimes DX \to Y \otimes DX,$$

where the second map is $h^a = Dg^b$. ■

We are now ready to prove Theorem 2.153. We will instead prove the following stronger result.

Theorem 2.163. Let X be a finite spectrum of type n. Then X admits a v_n-self map, and it is unique up to taking powers.

Proof. The uniqueness was already shown in Proposition 2.161. By looking at some explicit examples of finite spectra of type n (e.g., from lens spaces), it only remains to show that the subcategory of spectra of type at least n admitting a v_n-self map is a thick subcategory. It's not hard to show that admitting such a map is preserved by produces or retracts, so it only remains to show the claim for cofibers. But this follows by taking a cofiber of $f\colon X \to Y$ in the diagram in Corollary 2.162. ■

2.8.3 Bousfield Localization

Our next goal is to complete a spectrum. To this end, we will discuss the more general notion of Bousfield localization. We will need a size condition.

Definition 2.164 (presentable)**.** Fix a ring spectrum R. Then a full subcategory $\mathcal{C} \subseteq \mathrm{Mod}(R)$ is *presentable* if and only if it satisfies the following.

- Closure: $\mathcal{C}$ is closed under de-suspensions and all colimits.
- Size: there is a set of objects generating $\mathcal{C}$ under colimits.

Here is the main theorem; it is an application of some Adjoint functor theorem.

Theorem 2.165. Let R be a ring spectrum, and let $\mathcal{C} \subseteq \mathrm{Mod}(R)$ be a presentable full subcategory. Then the inclusion $\mathcal{C} \to \mathrm{Mod}(R)$ preserves colimits and admits a right adjoint G given by

$$G(M) := \operatorname*{colim}_{\substack{N \in \mathcal{C} \\ f\colon N \to M}} N.$$

Remark 2.166. We have tended to ignore set-theoretic conditions, but this one is important because we are trying to apply an Adjoint functor theorem. In particular, the colimit does not even really make sense if $\mathcal{C}$ is not even generated by a set of objects.

Theorem 2.165 gives rise to our localization.

Definition 2.167 (Bousfield localization)**.** Let $\mathcal{C}$ be a presentable full subcategory of $\mathrm{Mod}(R)$, where R is a ring spectrum, and let $G_{\mathcal{C}}$ be the right adjoint of the inclusion. For an R-module M, the cofiber of the unit map $G_{\mathcal{C}}M \to M$ is the *Bousfield localization*, labeled $L_{\mathcal{C}}M$.

Example 2.168. Let $\mathcal{C}$ denote the collection of $\mathbb{S}_{(p)}$-modules generated under all colimits from $\mathbb{S}/p$ and its de-suspensions. Then what is left turns out to be $L_{\mathcal{C}}X = X[1/p]$. Thus, Bousfield localization recovers usual localization.

Remark 2.169. In fact, we can use Theorem 2.165 to show that $R[1/r]$ is an $\mathbb{E}_\infty$-ring for any $r \in \pi_* R$, essentially by identifying its category of modules.

Remarkably, we can also recover completion.

Example 2.170. Let $\mathcal{C}$ denote the collection of $\mathbb{S}_{(p)}$-modules for which the canonical map $M \to M[1/p]$. (It is not obvious that this subcategory is generated by a set of objects!) Then $L_{\mathcal{C}}X$ is the completion $X_p^\wedge$.

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